Update étude des données

05 mai 2026
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ISAE-SUPAERO / ENAC / Edge Spectrum

Sommaire

- Equipment Monitoring :
  - Contributions individuelles des features
  - Détection d'anomalies
- Datasets Pronostia :
  - Extraction features
  - Trajectoire de dégradation
  - Evaluation scenario by_condition
- Datasets CWRU :
  - Trajectoire de dégradation
  - Evaluation scenario by_fault_type
  - Evaluation scenario by_severity

Equipment Monitoring _ Contributions individuelles des features

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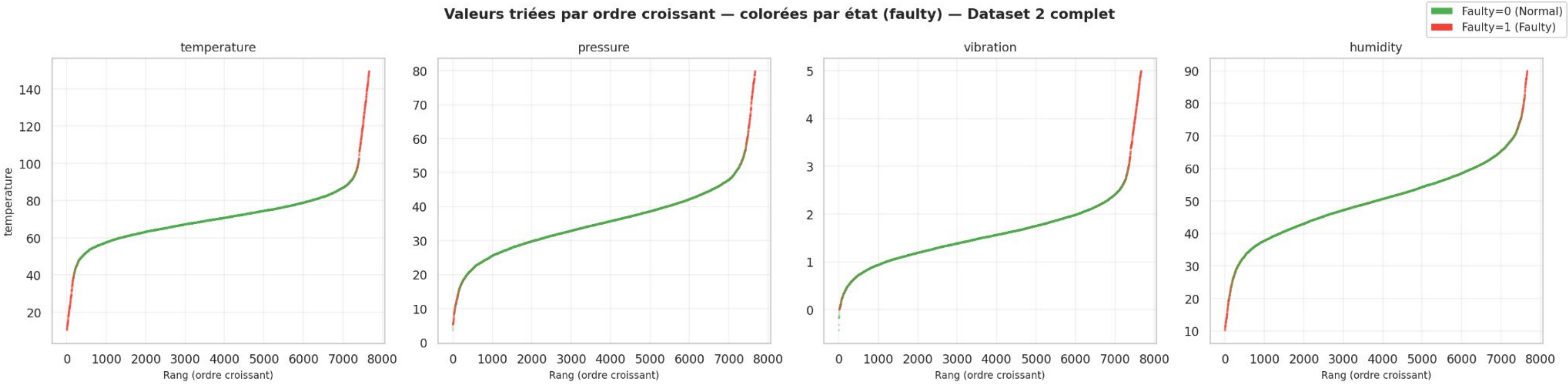

Equipment Monitoring _ Contributions individuelles des features

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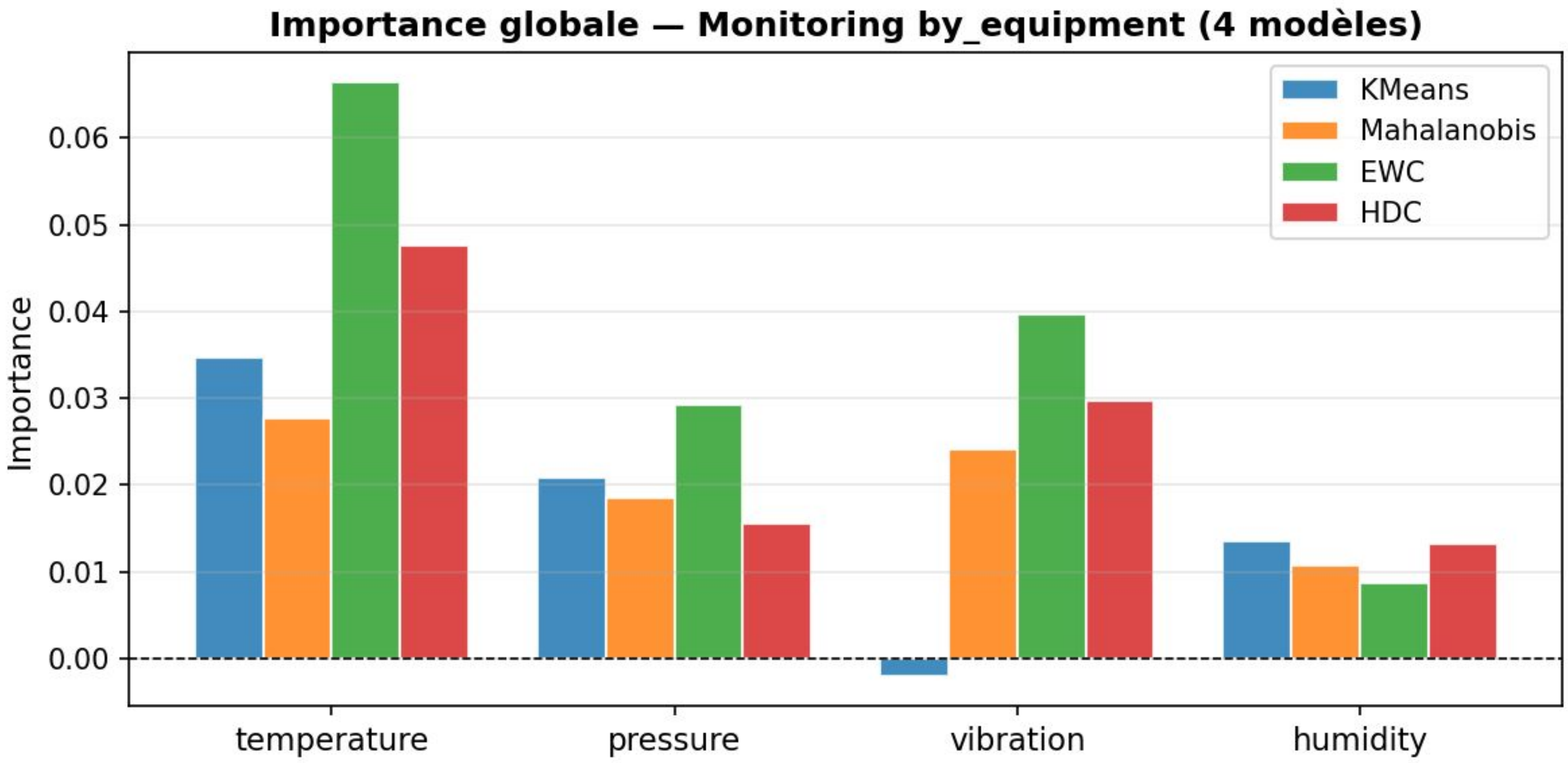
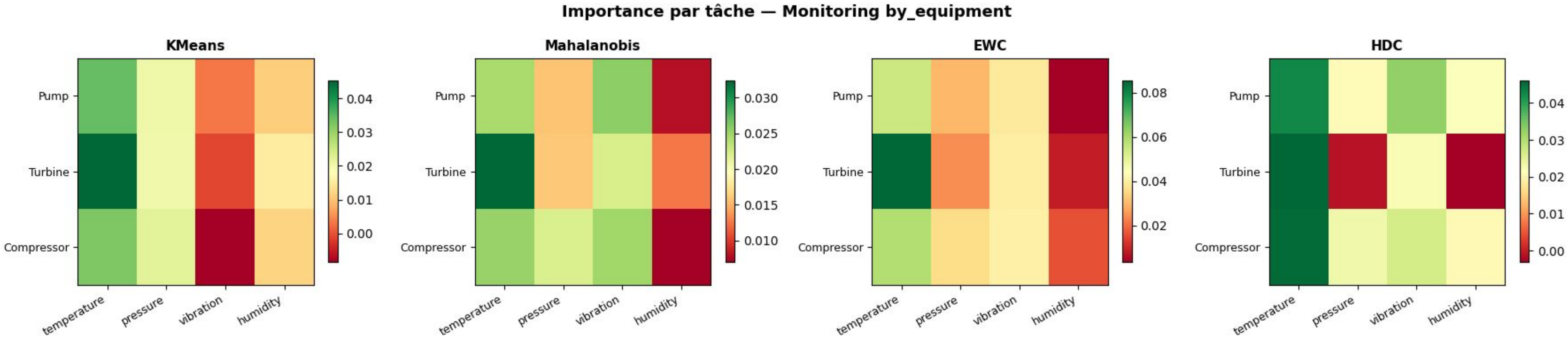
Permutation Importance
pour chaque feature, permuter sa colonne

Feature Masking
remplace toute la colonne j par 0.0

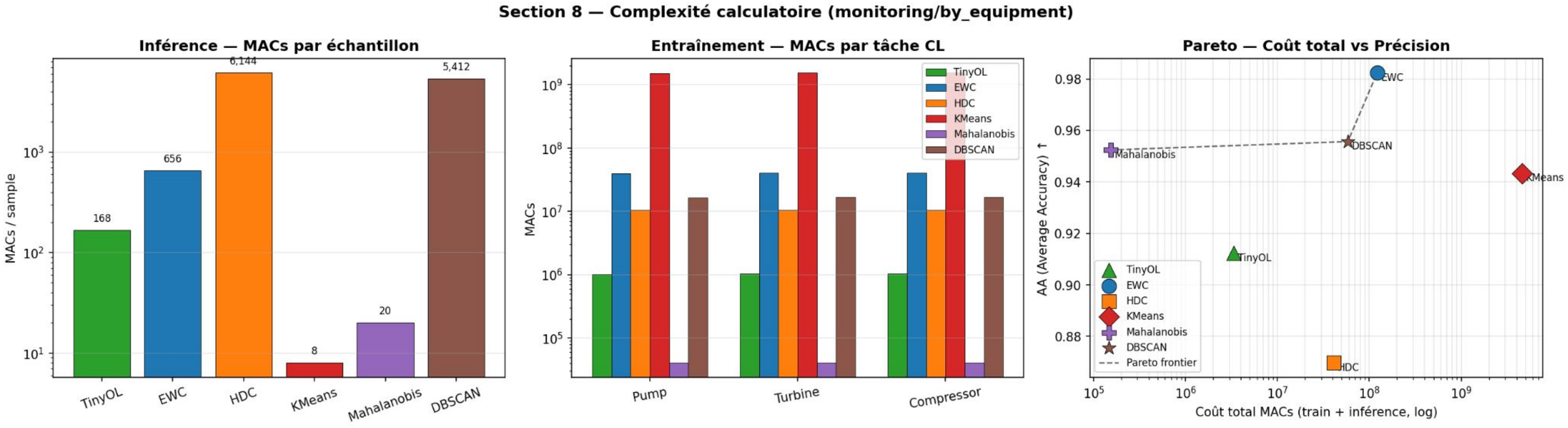
Equipment Monitoring _ Contributions individuelles des features (equipment)



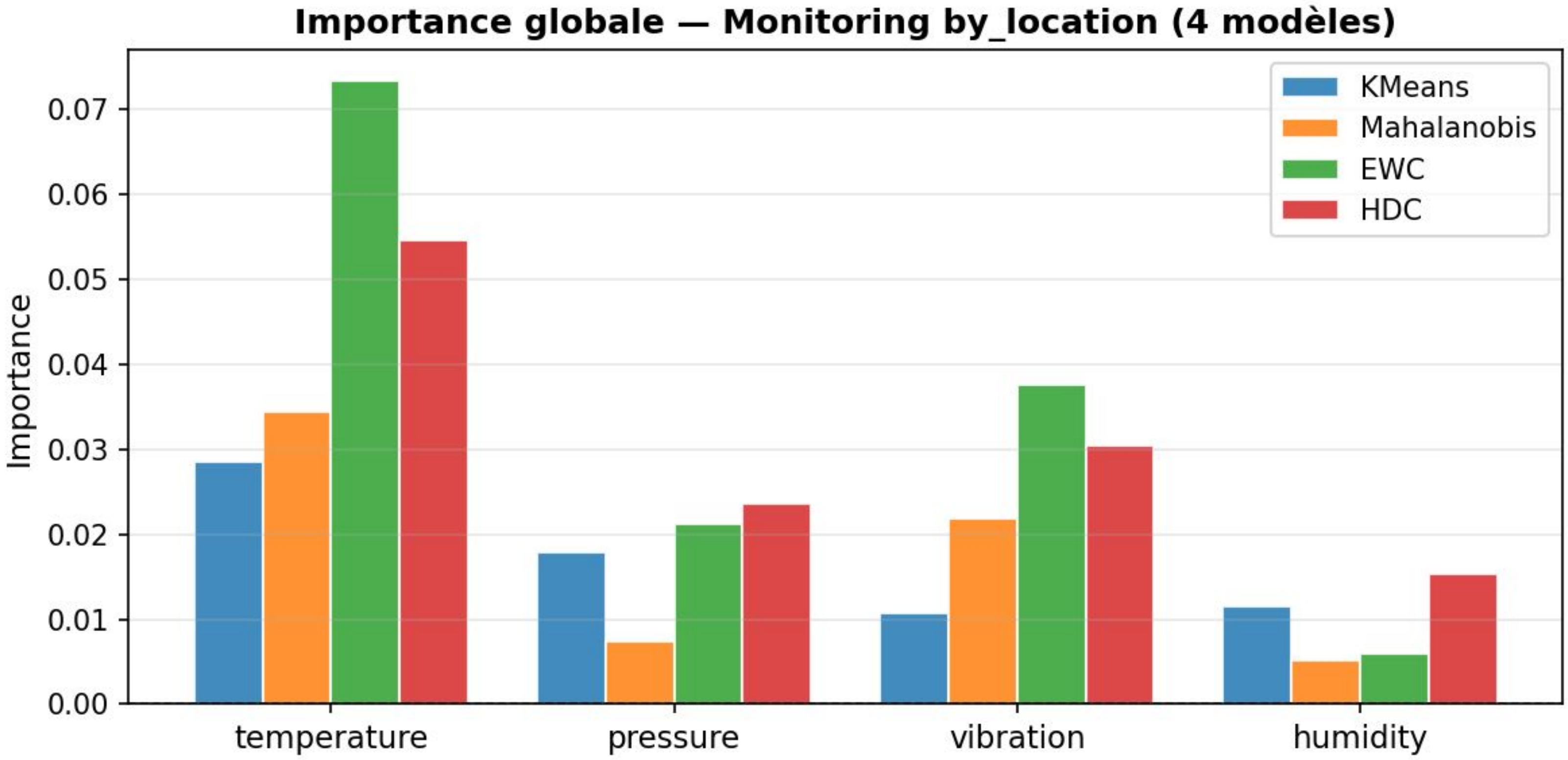
Equipment Monitoring _ Contributions individuelles des features (equipment)



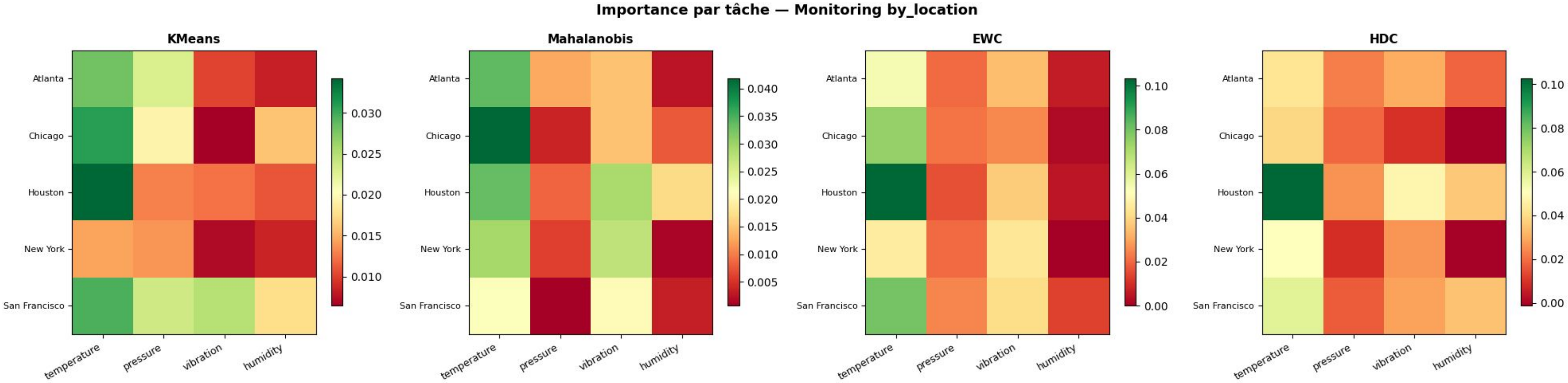
Analyse complexité monitoring by_equipment



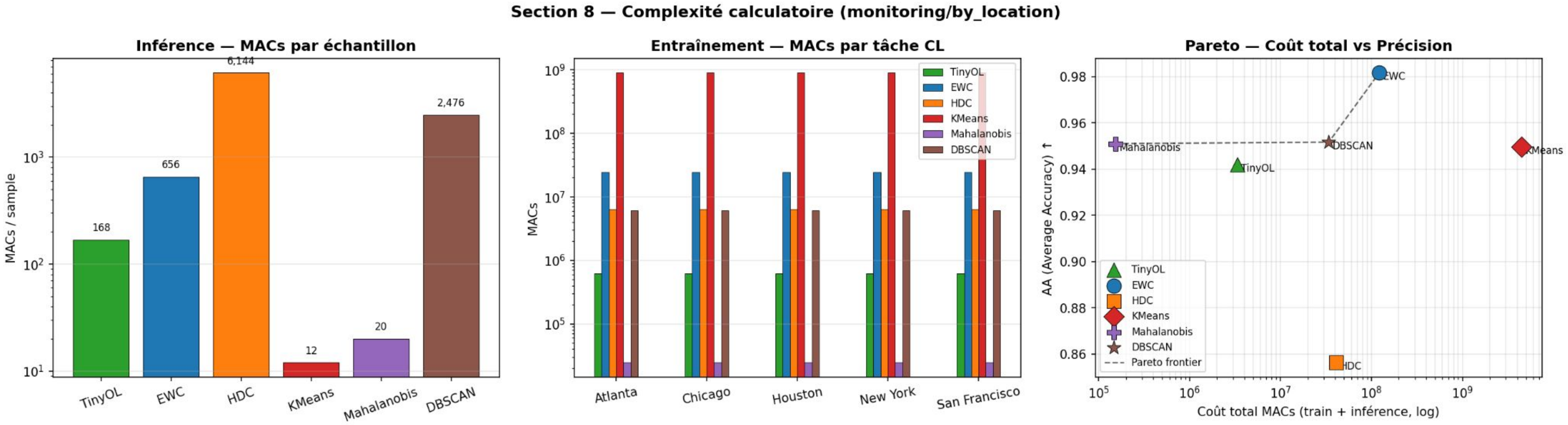
Equipment Monitoring _ Contributions individuelles des features (location)



Equipment Monitoring _ Contributions individuelles des features (location)



Analyse complexité monitoring by_location



Equipment Monitoring _ détection d'anomalies

modèle s'entraîne uniquement sur les échantillons normaux

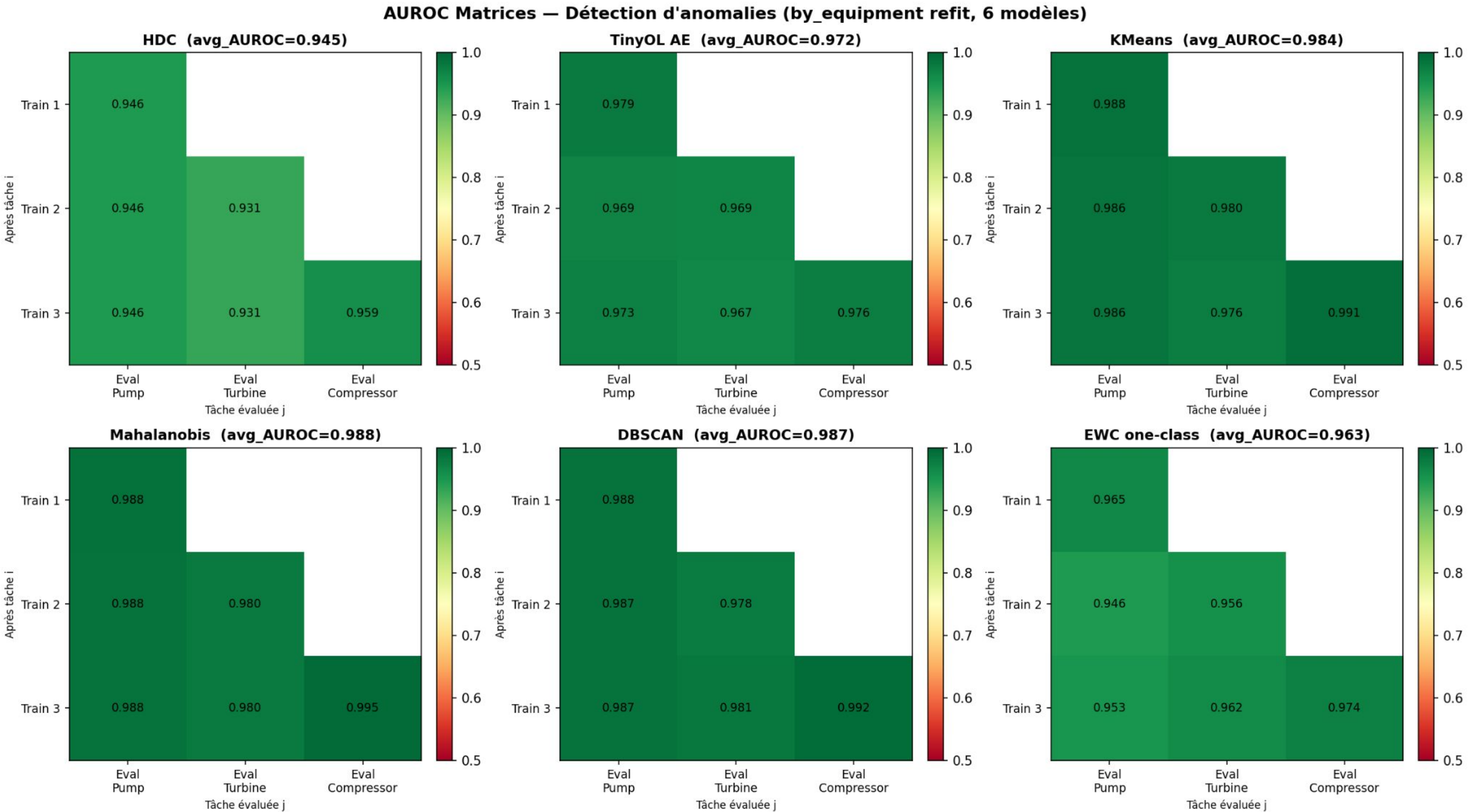
détecte les anomalies à l'inférence

le modèle réapprend depuis zéro à chaque nouvelle tâche → test de l'oubli catastrophique maximal

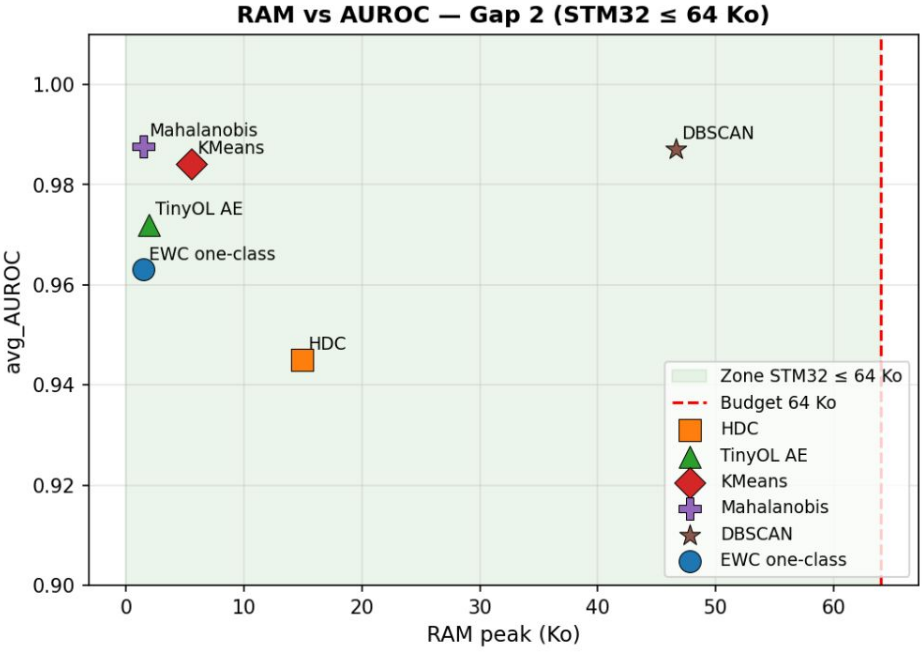
Equipment Monitoring _ by_equipment, stratégie refit

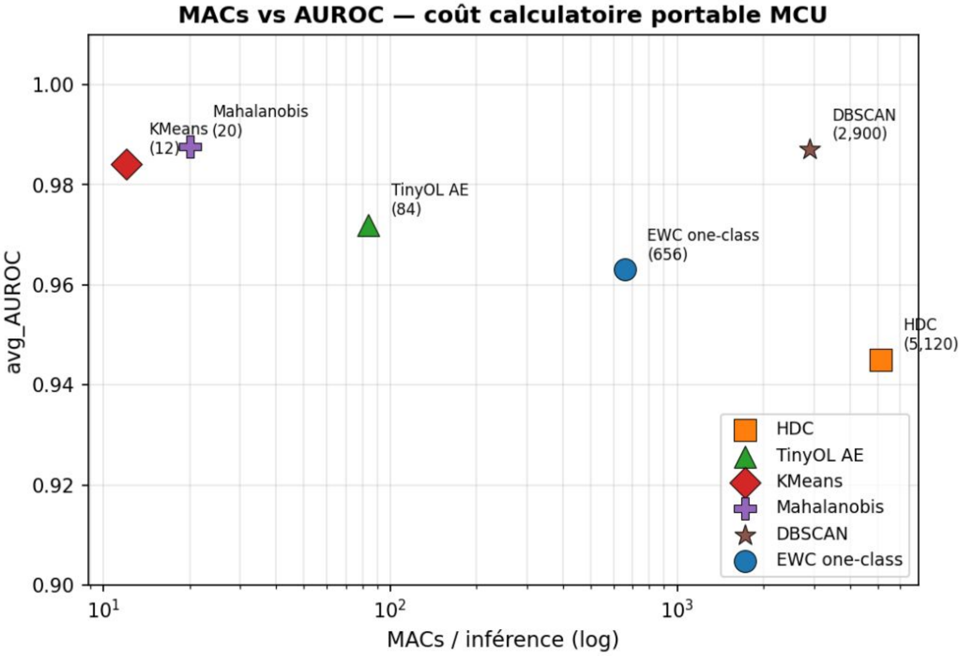
Scénario by_equipment, stratégie refit
(3 tâches : Pump → Turbine → Compressor)

Equipment Monitoring _ by_equipment, stratégie refit

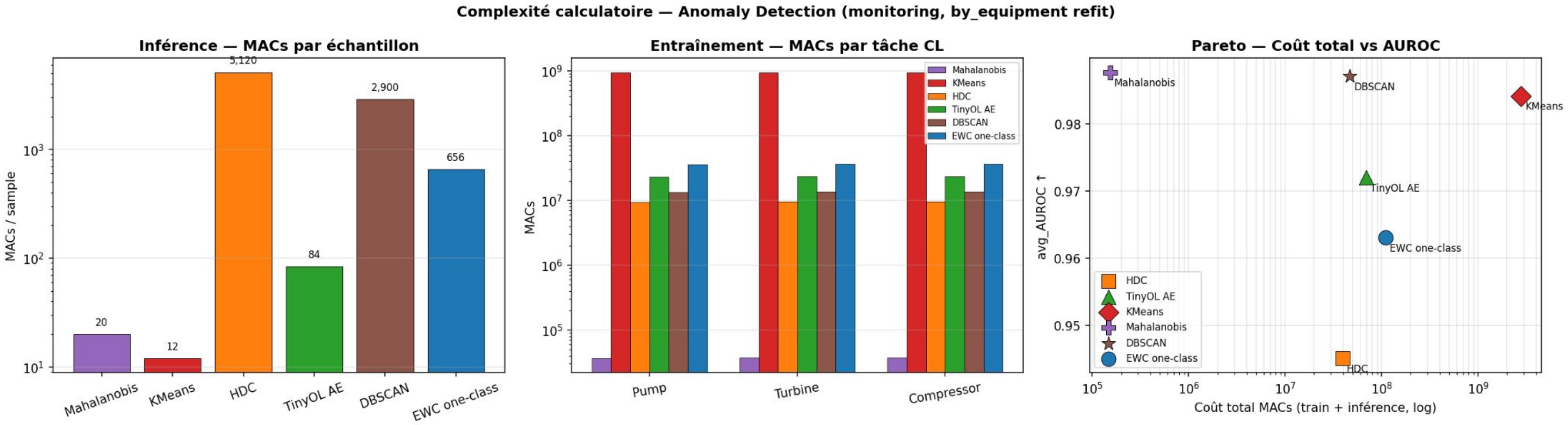


Equipment Monitoring _ by_equipment, stratégie refit



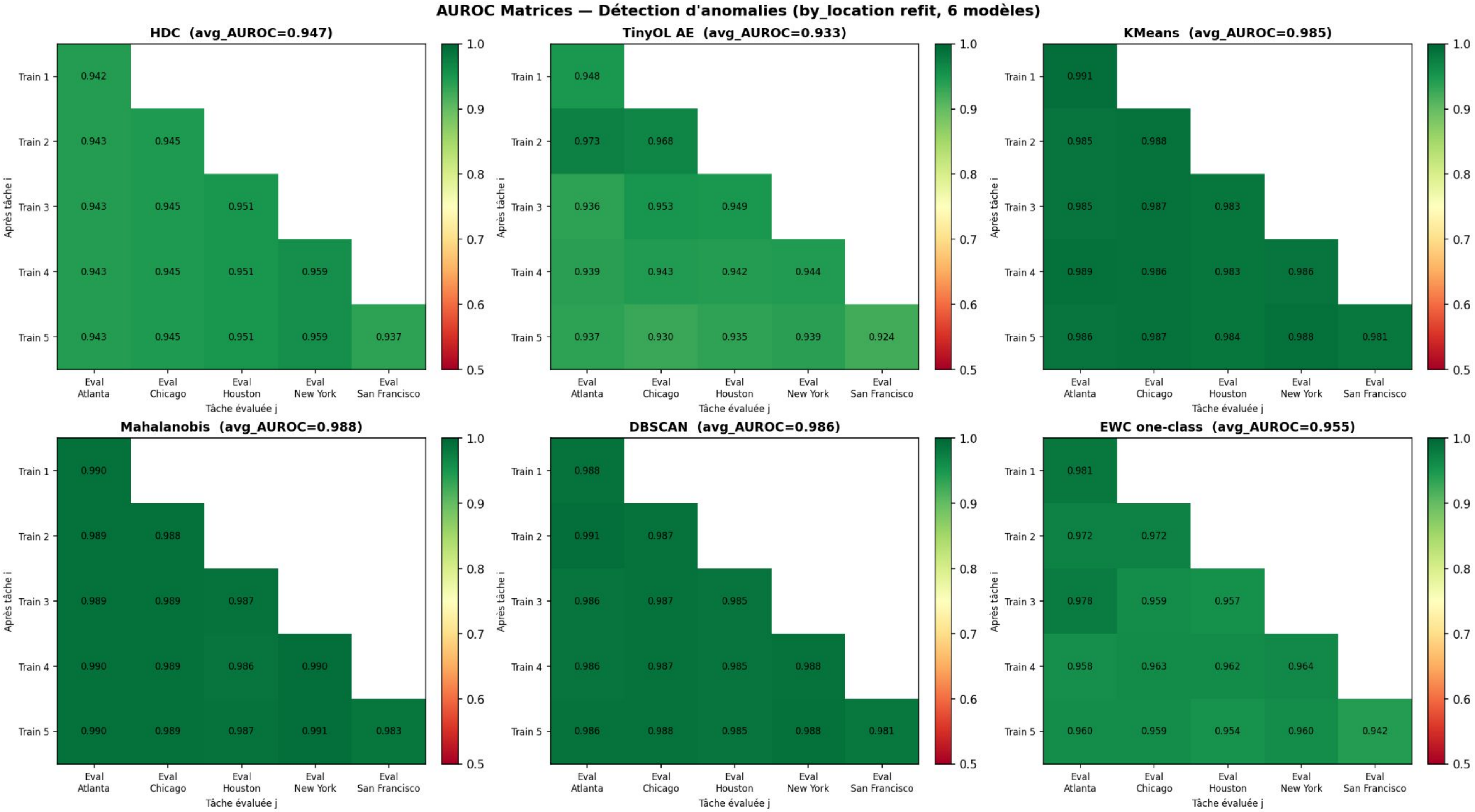


Equipment Monitoring _ by_equipment, stratégie refit



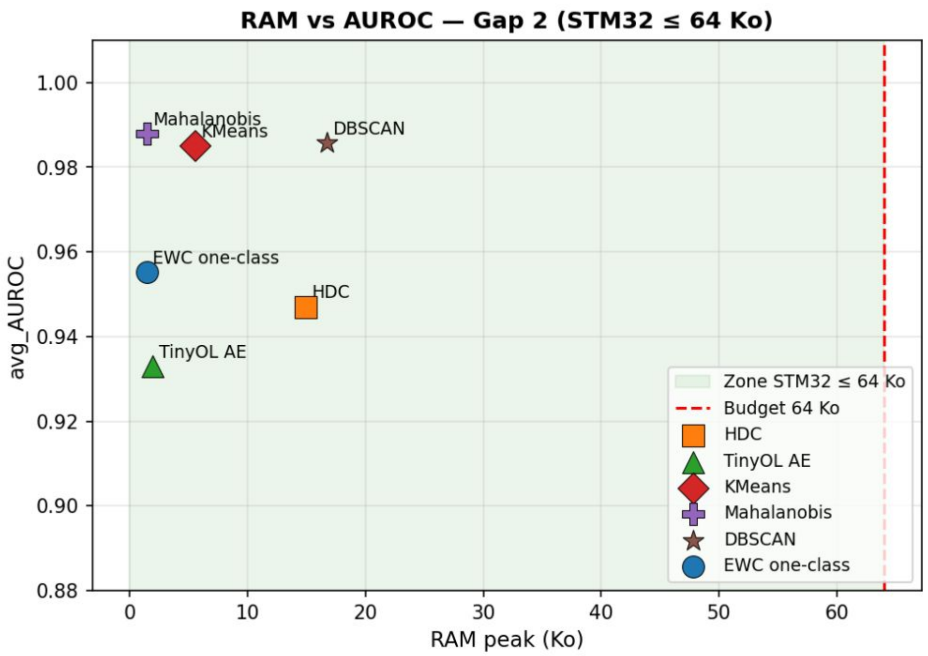
Equipment Monitoring _ by_location, stratégie refit

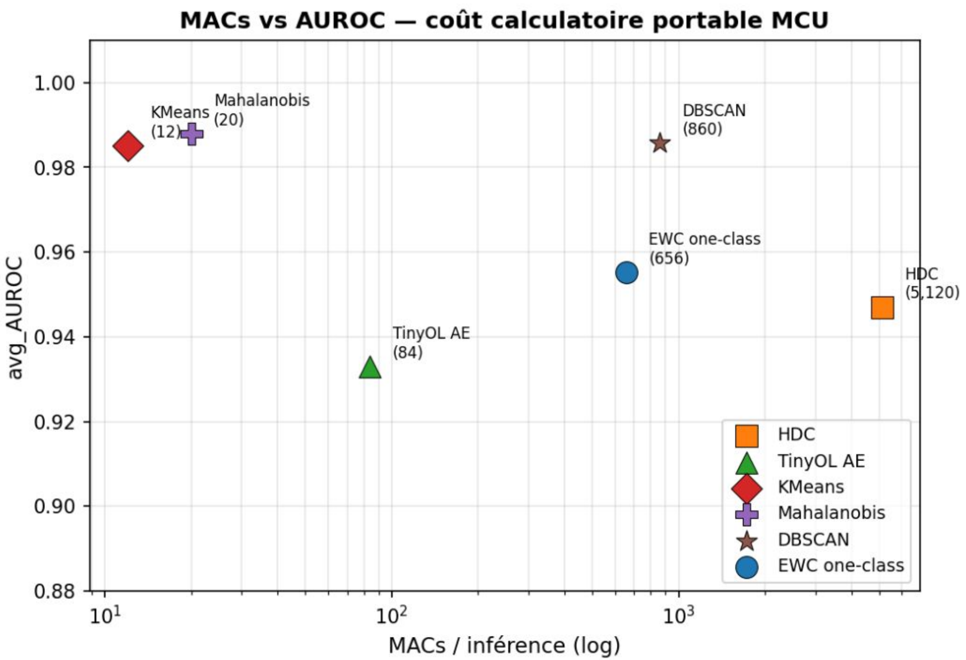
Scénario by_location, stratégie refit
(5 tâches : Atlanta → Chicago → Houston → New York → San Francisco)



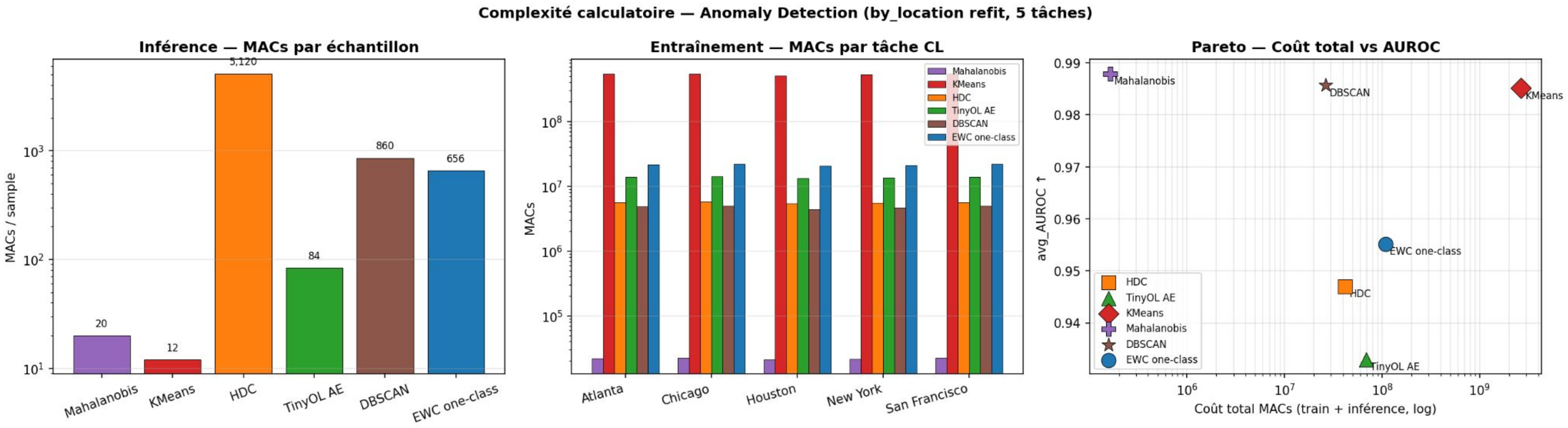
AUROC Matrices — Détection d'anomalies (by_location refit, 6 modèles)
HDC (avg_AUROC=0.947)
TinyOL AE (avg_AUROC=0.933)
KMeans (avg_AUROC=0.985)
Mahalanobis (avg_AUROC=0.988)
DBSCAN (avg_AUROC=0.986)
EWC one-class (avg_AUROC=0.955)
Après tâche i
Tâche évaluée j
Train 1
Train 2
Train 3
Train 4
Train 5
Eval Atlanta
Eval Chicago
Eval Houston
Eval New York
Eval San Francisco

Equipment Monitoring _ by_location, stratégie refit





Equipment Monitoring _ by_location, stratégie refit



Equipment Monitoring _

Datasets Pronostia

Experimental Platform for Bearings Accelerated Life Test

Roulements à billes soumis à des conditions : Charge et vitesse jusqu'à défaillance complète.

Chaque essai couvre un cycle de vie complet (sain -> défectueux)

Datasets Pronostia

2 capteurs vibratoires:

- Accéléromètre Horizontal et Vertical

Enregistre 0.1s tt 10s

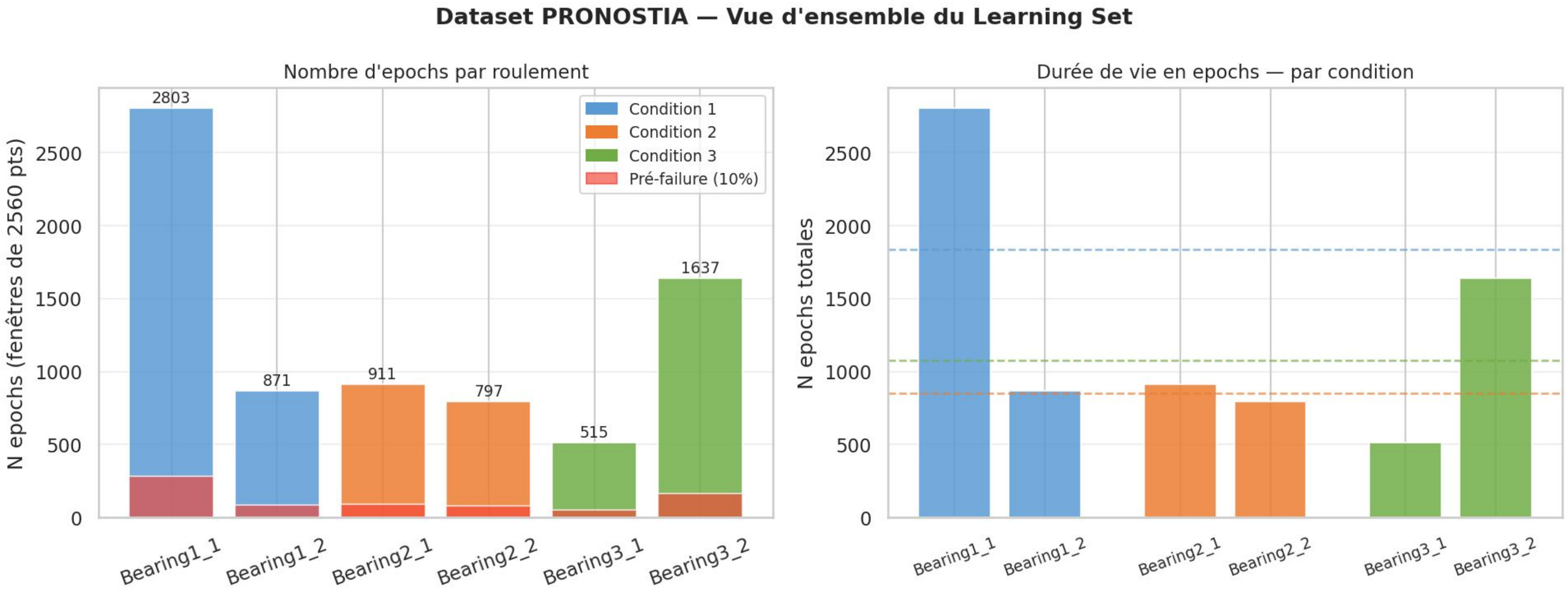
4 entrées temporelles:
- Heure
- Minute
- Seconde
- Microseconde

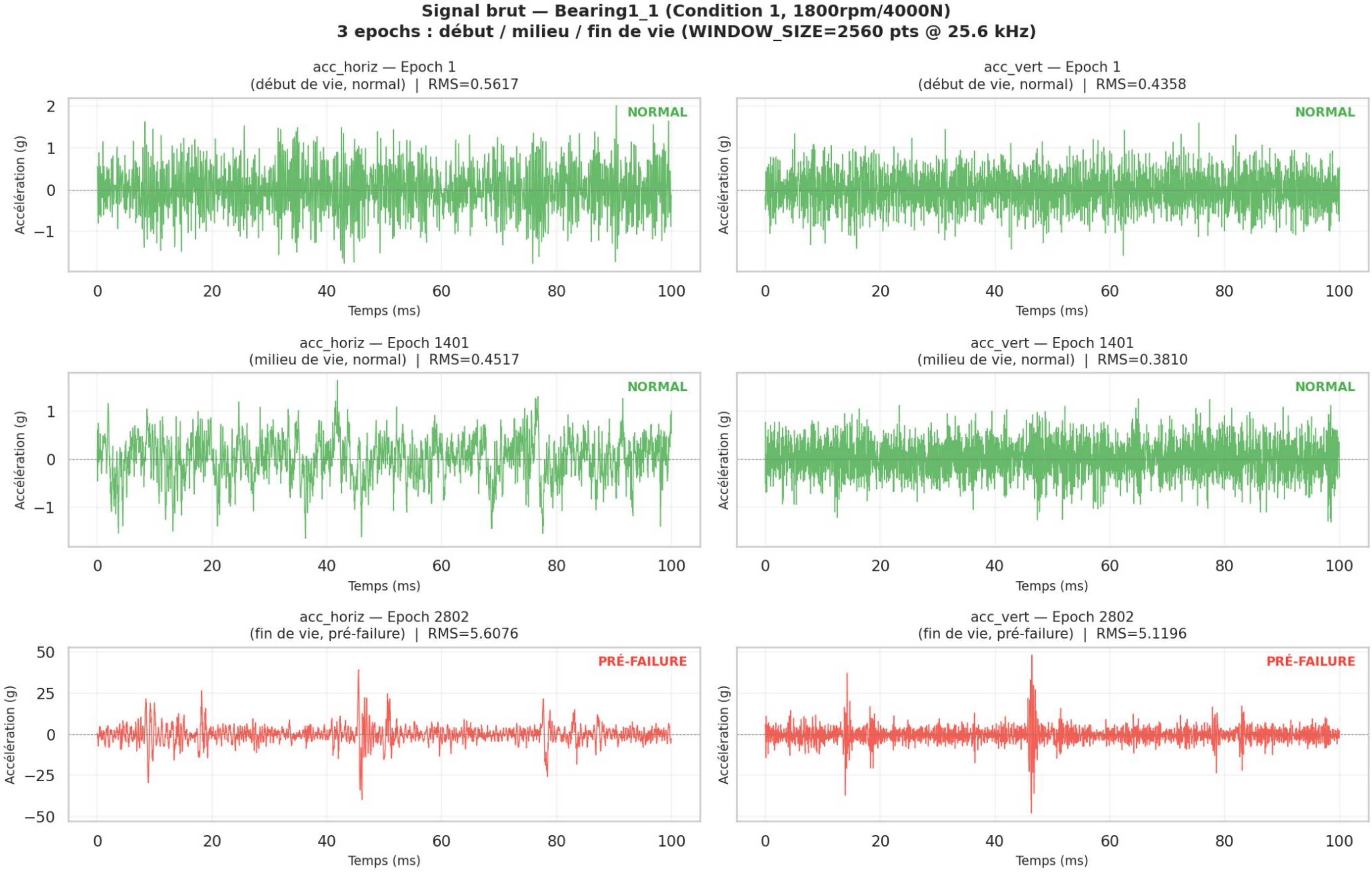
Column	**1**	**2**	**3**	**4**	**5**	**6**
Vibr. signal	Hour	Minute	Second	μ-second	Horiz. accel.	vert. accel.
Temp. signal	Hour	Minute	Second	0.x second	Rtd sensor	

Datasets Pronostia

Roulement	Condition	RPM	Charge (N)	Samples bruts	N epochs	N pré-failure	Taux failure
Bearing1_1	1	1800	4000	7175680	2803	281	10.0%
Bearing1_2	1	1800	4000	2229760	871	88	10.1%
Bearing2_1	2	1650	4200	2332160	911	92	10.1%
Bearing2_2	2	1650	4200	2040320	797	80	10.0%
Bearing3_1	3	1500	5000	1318400	515	52	10.1%
Bearing3_2	3	1500	5000	4190720	1637	164	10.0%

Datasets Pronostia




Signal brut — Bearing1_1 (Condition 1, 1800rpm/4000N)
3 epochs : début / milieu / fin de vie (WINDOW_SIZE=2560 pts @ 25.6 kHz)
acc_horiz — Epoch 1
(début de vie, normal) | RMS=0.5617
NORMAL
acc_vert — Epoch 1
(début de vie, normal) | RMS=0.4358
NORMAL
acc_horiz — Epoch 1401
(milieu de vie, normal) | RMS=0.4517
NORMAL
acc_vert — Epoch 1401
(milieu de vie, normal) | RMS=0.3810
NORMAL
acc_horiz — Epoch 2802
(fin de vie, pré-failure) | RMS=5.6076
PRÉ-FAILURE
acc_vert — Epoch 2802
(fin de vie, pré-failure) | RMS=5.1196
PRÉ-FAILURE
Accélération (g)
Temps (ms)

Datasets Pronostia _ Extraction de features

2 canaux vibratoires 13 entrées

signal en fenêtres de 2560 points:

- temporal_position = start / (n_total - window_size)

Datasets Pronostia _ Extraction de features

mean : le niveau moyen du signal sur la fenêtre.

std : la dispersion autour de la moyenne.

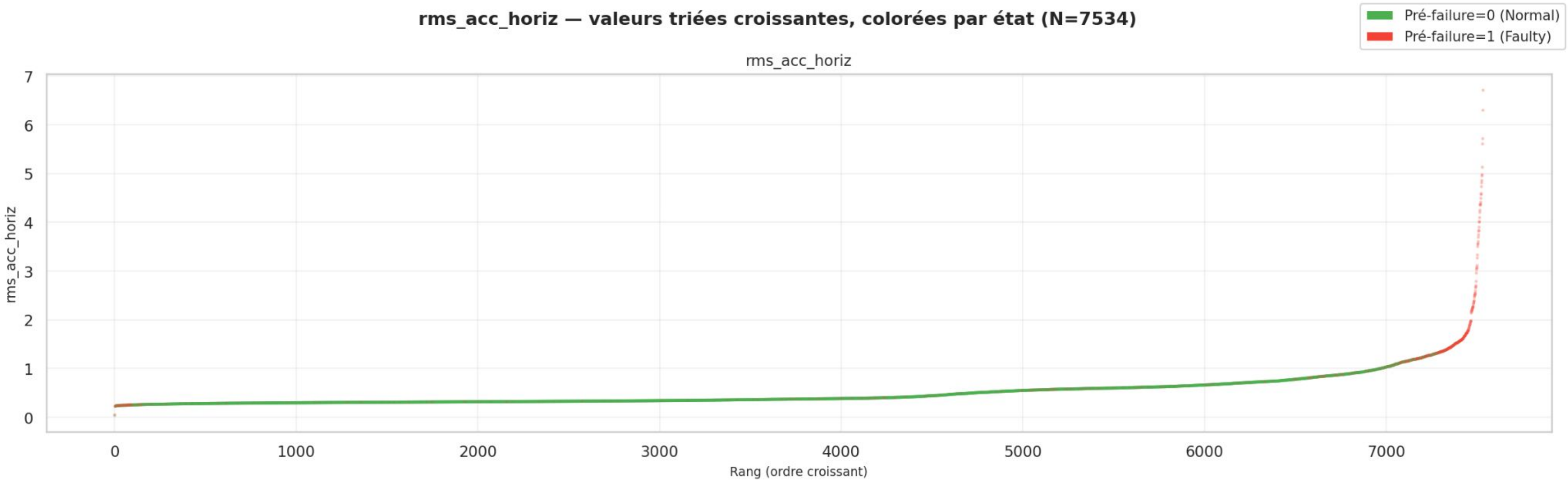
rms : le niveau d’énergie global du signal.

kurtosis : mesure à quel point le signal contient des pics rares et forts. I

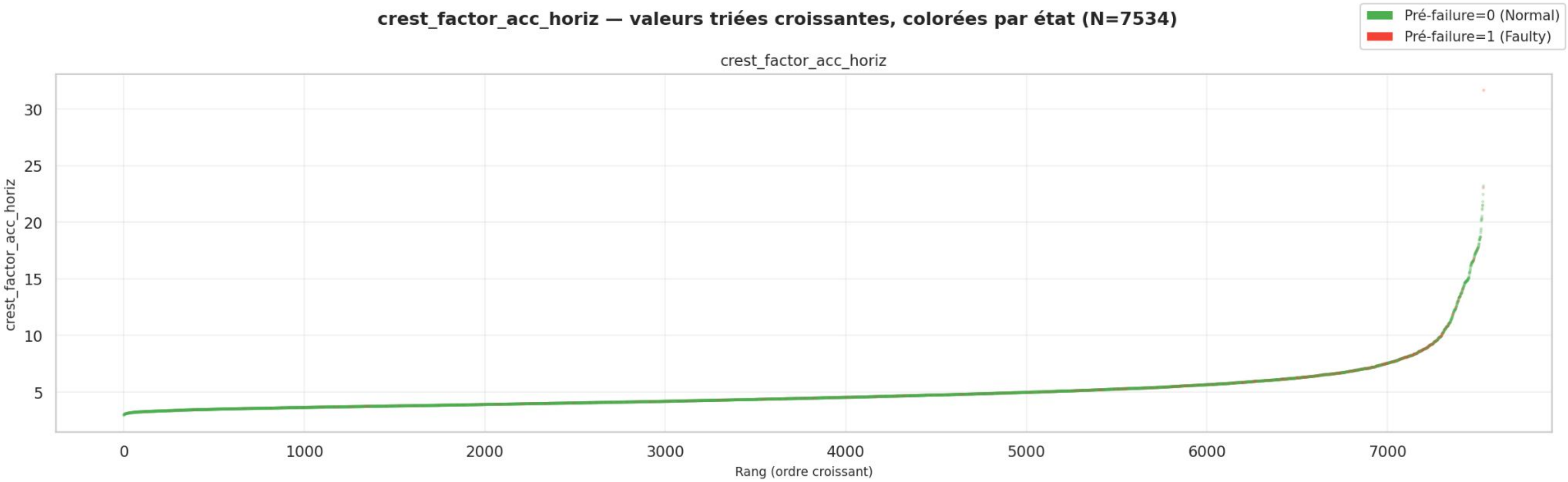
peak : la plus grande amplitude absolue observée dans la fenêtre.

crest_factor : compare le pic au niveau moyen d’énergie.

Datasets Pronostia _ Trajectoire de dégradation



Datasets Pronostia _ Trajectoire de dégradation



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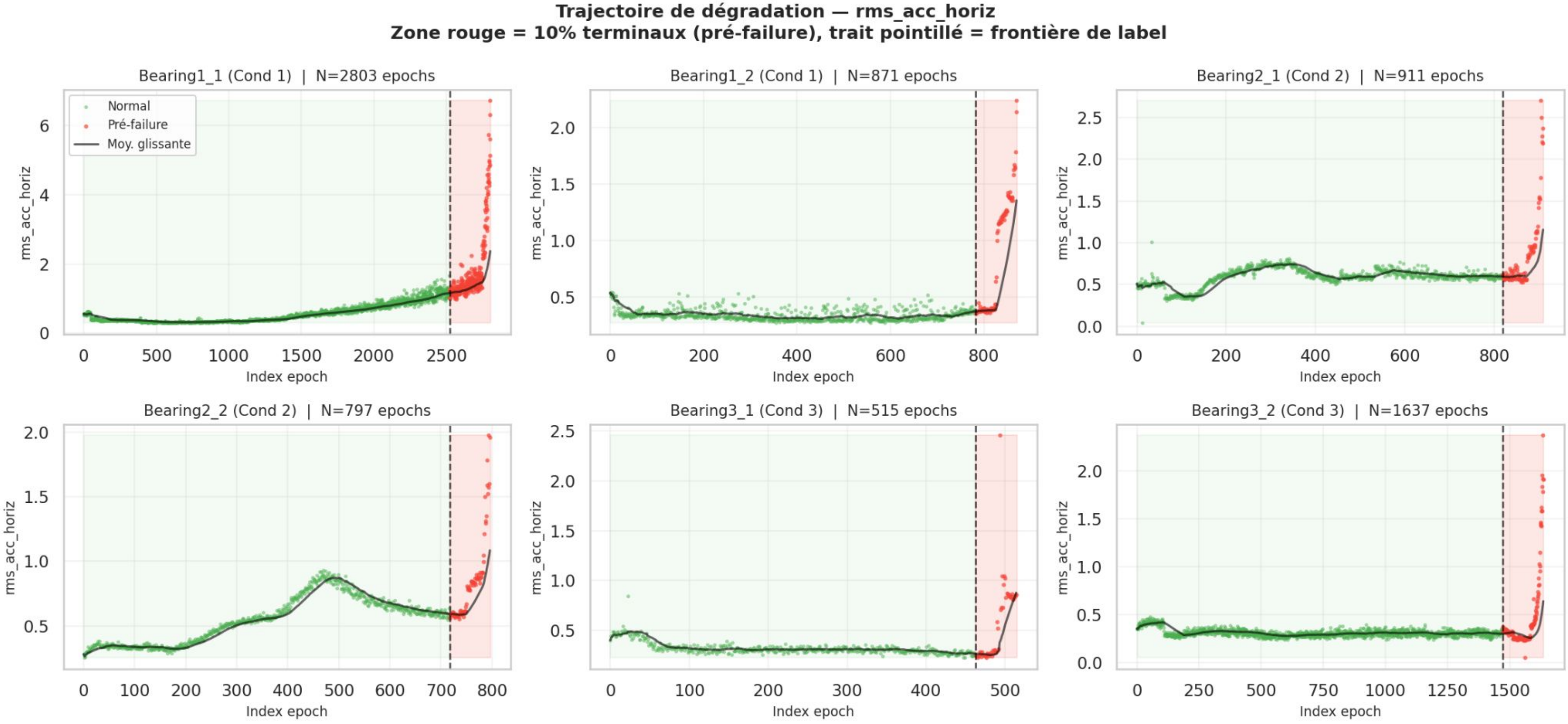
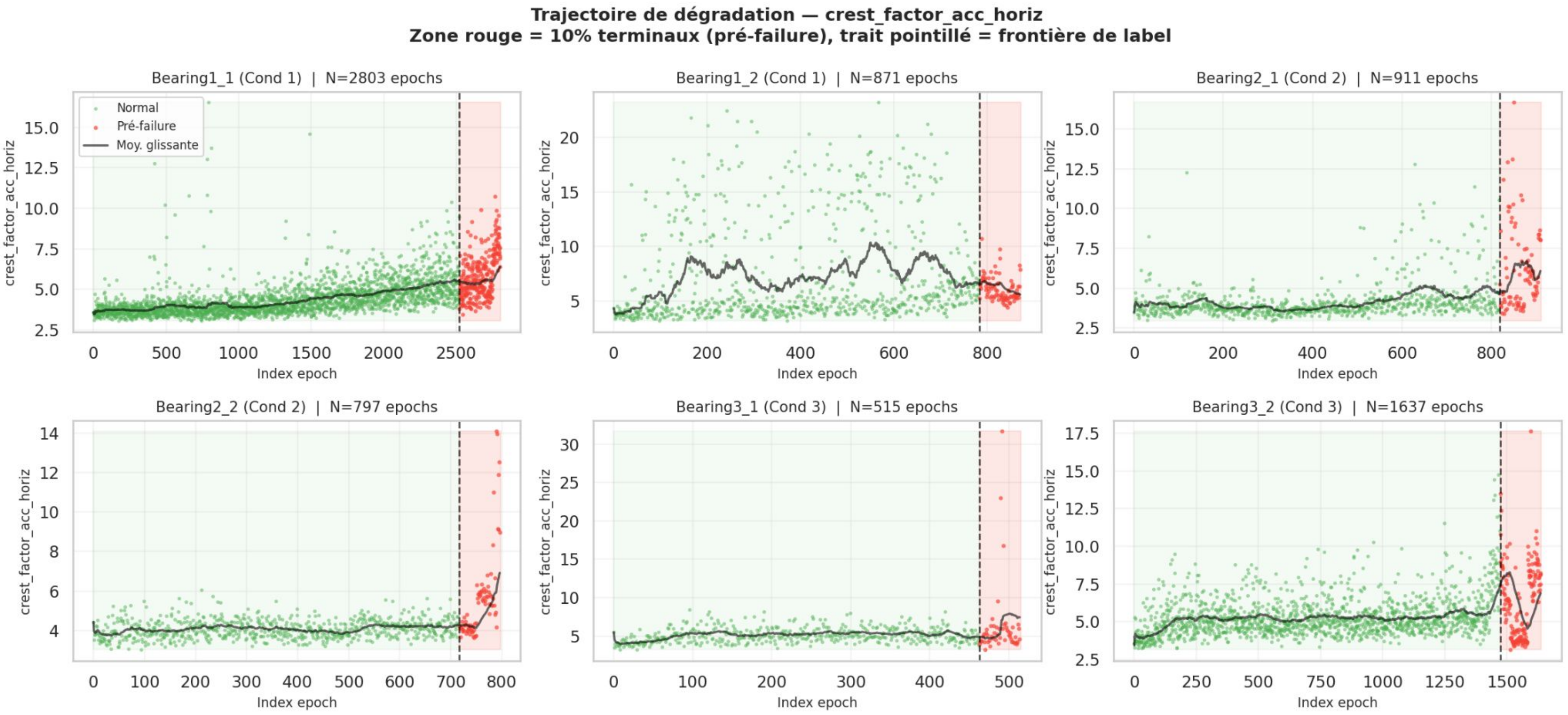
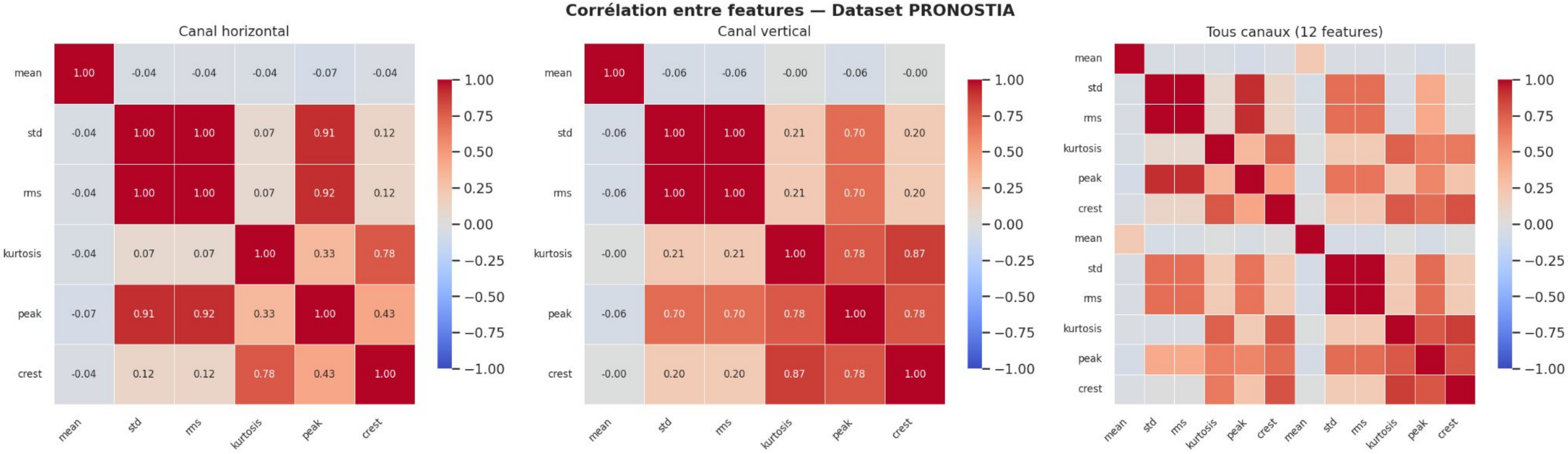
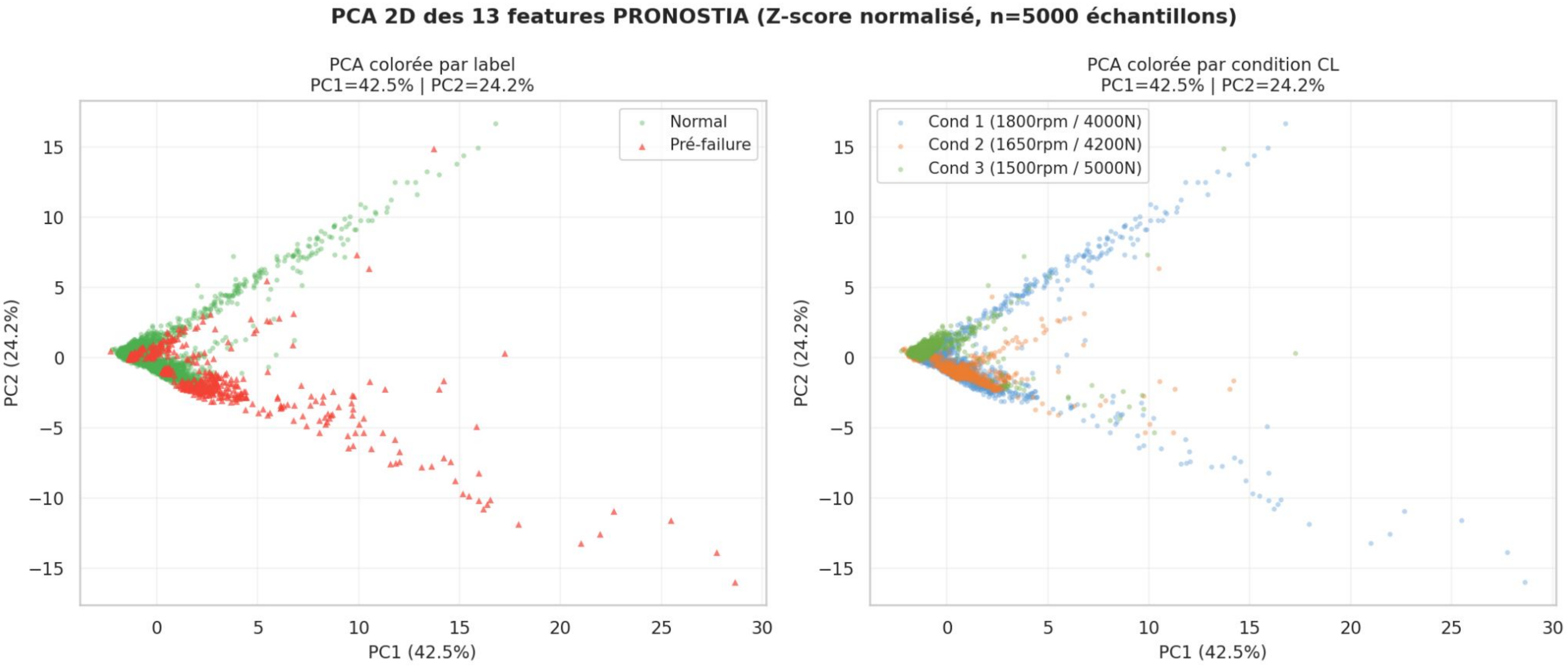


Datasets Pronostia _ Trajectoire de dégradation

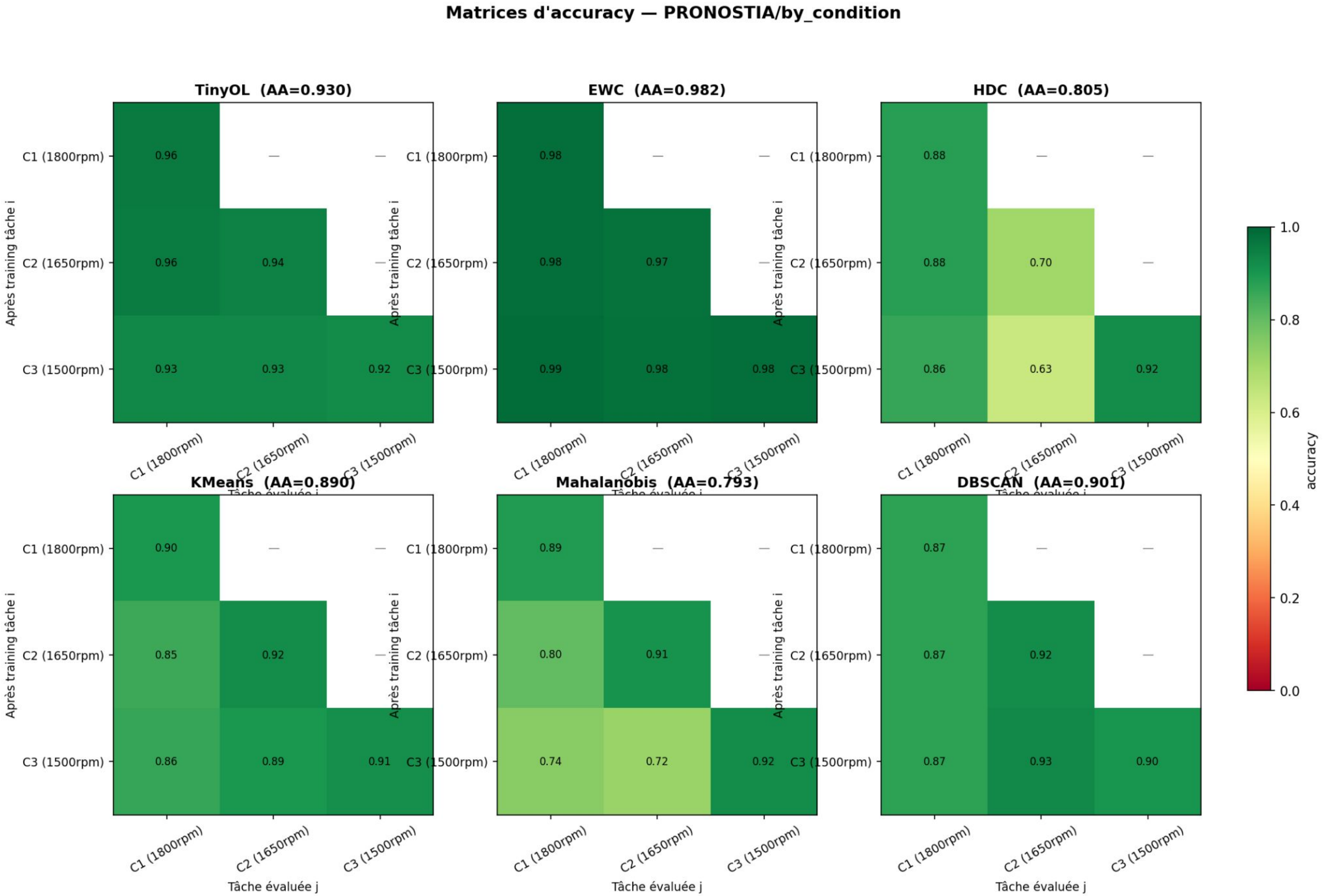


Datasets Pronostia

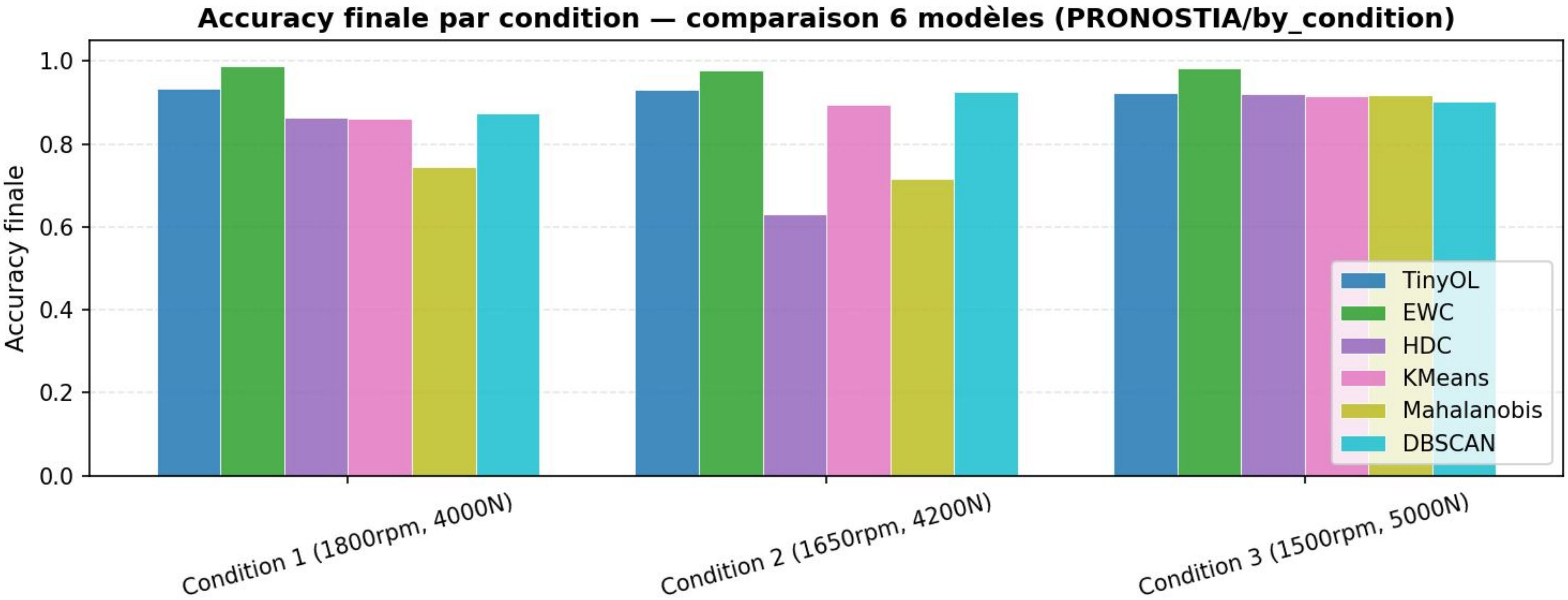


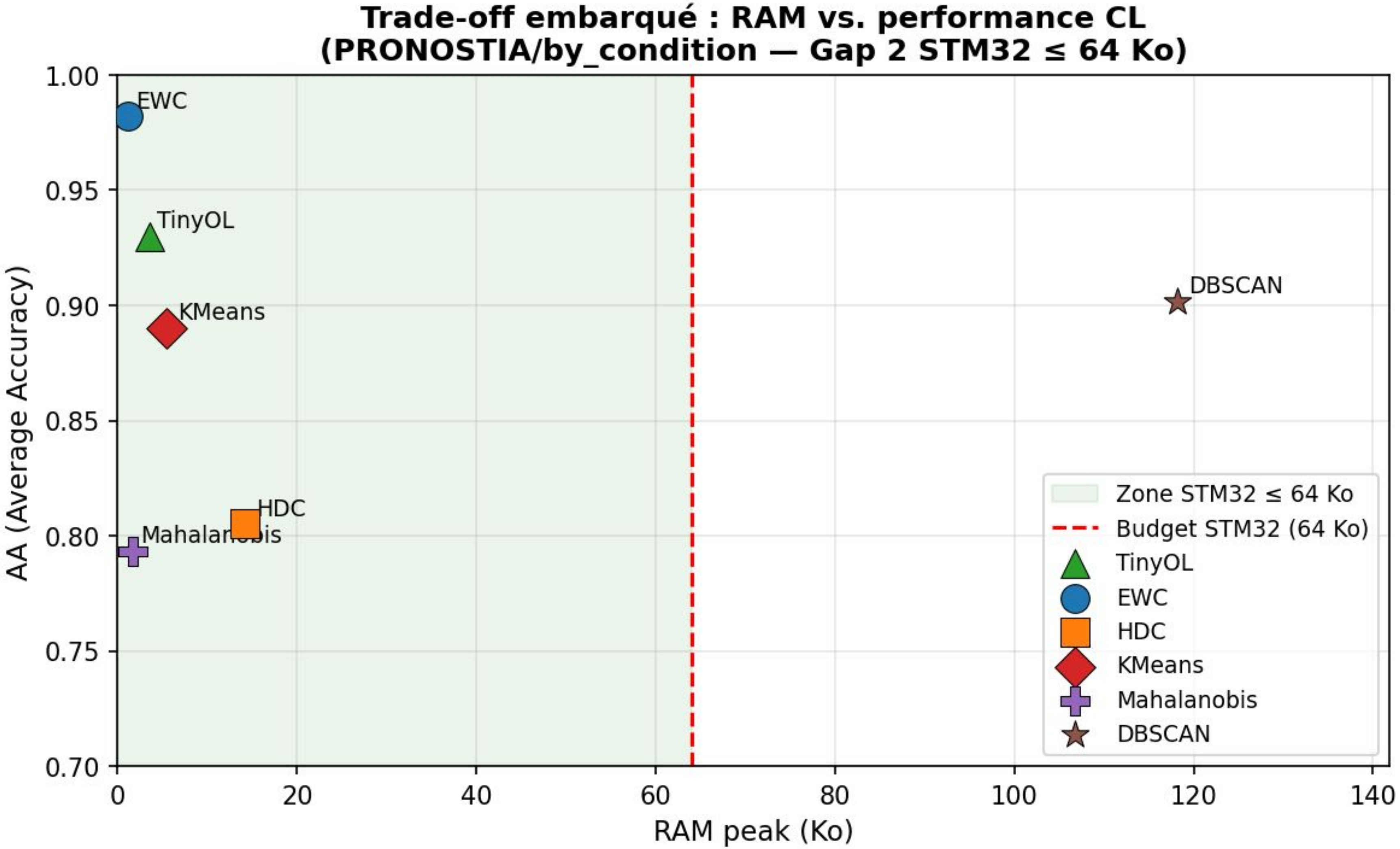

PCA 2D des 13 features PRONOSTIA (Z-score normalisé, n=5000 échantillons)
PCA colorée par label
PC1=42.5% | PC2=24.2%
Normal
Pré-failure
PC1 (42.5%)
PC2 (24.2%)
PCA colorée par condition CL
PC1=42.5% | PC2=24.2%
Cond 1 (1800rpm / 4000N)
Cond 2 (1650rpm / 4200N)
Cond 3 (1500rpm / 5000N)
PC1 (42.5%)
PC2 (24.2%)

Datasets Pronostia

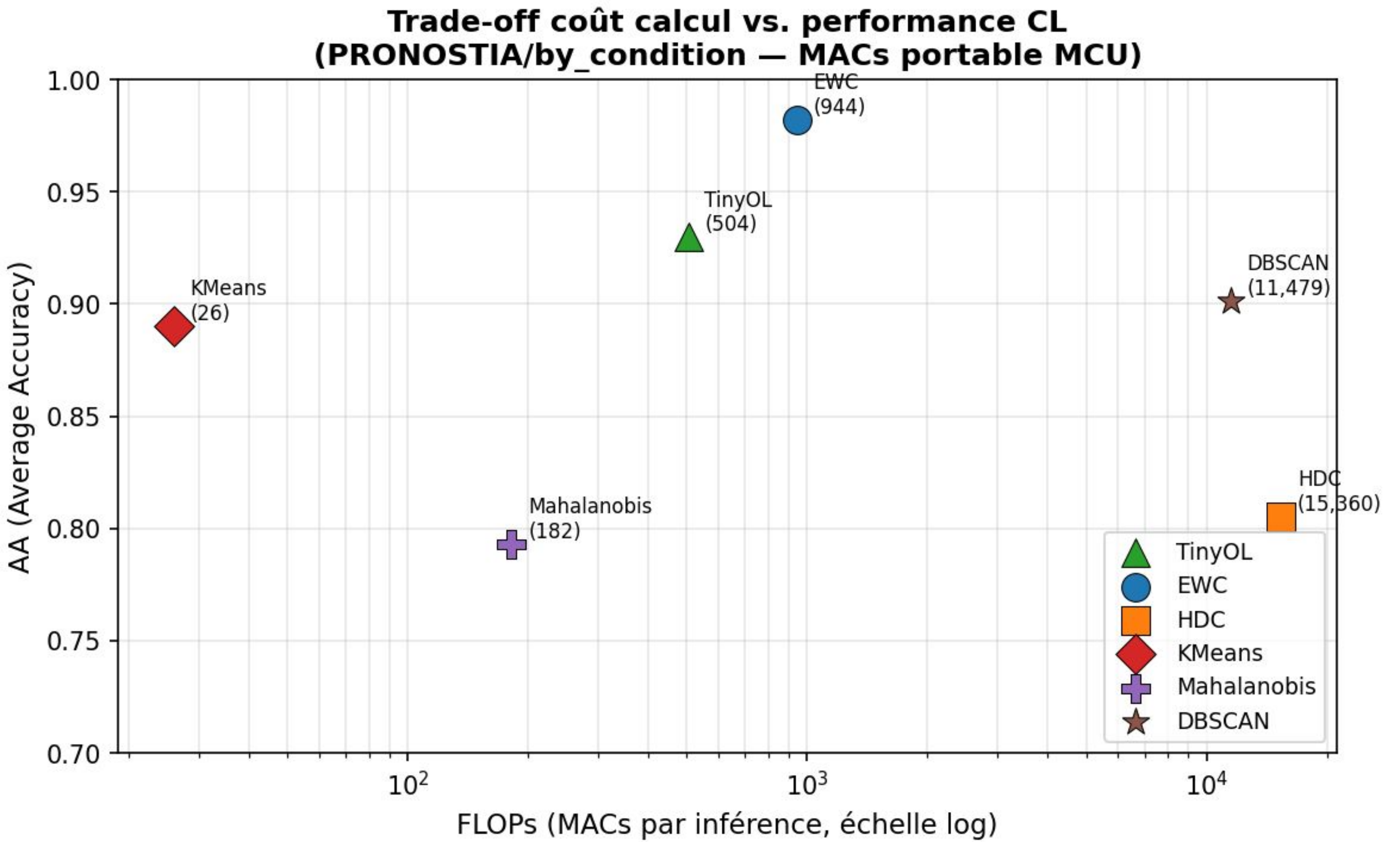

Matrices d'accuracy — PRONOSTIA/by_condition
TinyOL (AA=0.930)
EWC (AA=0.982)
HDC (AA=0.805)
KMeans (AA=0.890)
Mahalanobis (AA=0.793)
DBSCAN (AA=0.901)
Après training tâche i
Tâche évaluée j
C1 (1800rpm)
C2 (1650rpm)
C3 (1500rpm)
accuracy

Datasets Pronostia _ pronostia by_condition

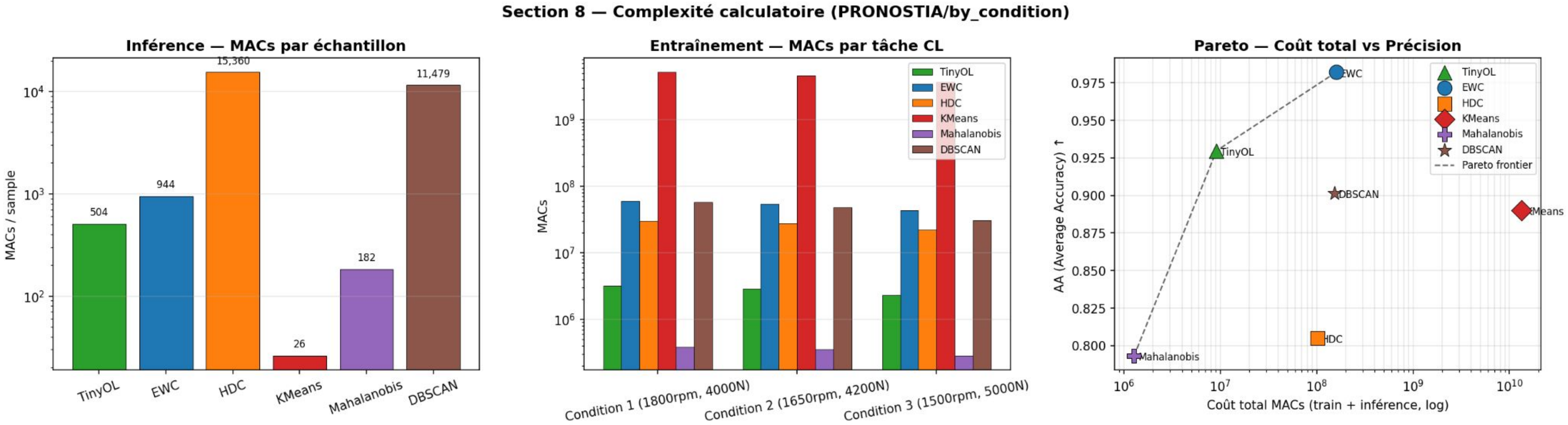



Trade-off embarqué : RAM vs. performance CL
(PRONOSTIA/by_condition — Gap 2 STM32 ≤ 64 Ko)
AA (Average Accuracy)
RAM peak (Ko)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0
20
40
60
80
100
120
140
EWC
TinyOL
KMeans
HDC
Mahalanobis
DBSCAN
Zone STM32 ≤ 64 Ko
Budget STM32 (64 Ko)
TinyOL
EWC
HDC
KMeans
Mahalanobis
DBSCAN

Datasets Pronostia _ pronostia by_condition



Datasets Pronostia _ pronostia by_condition



Datasets Pronostia

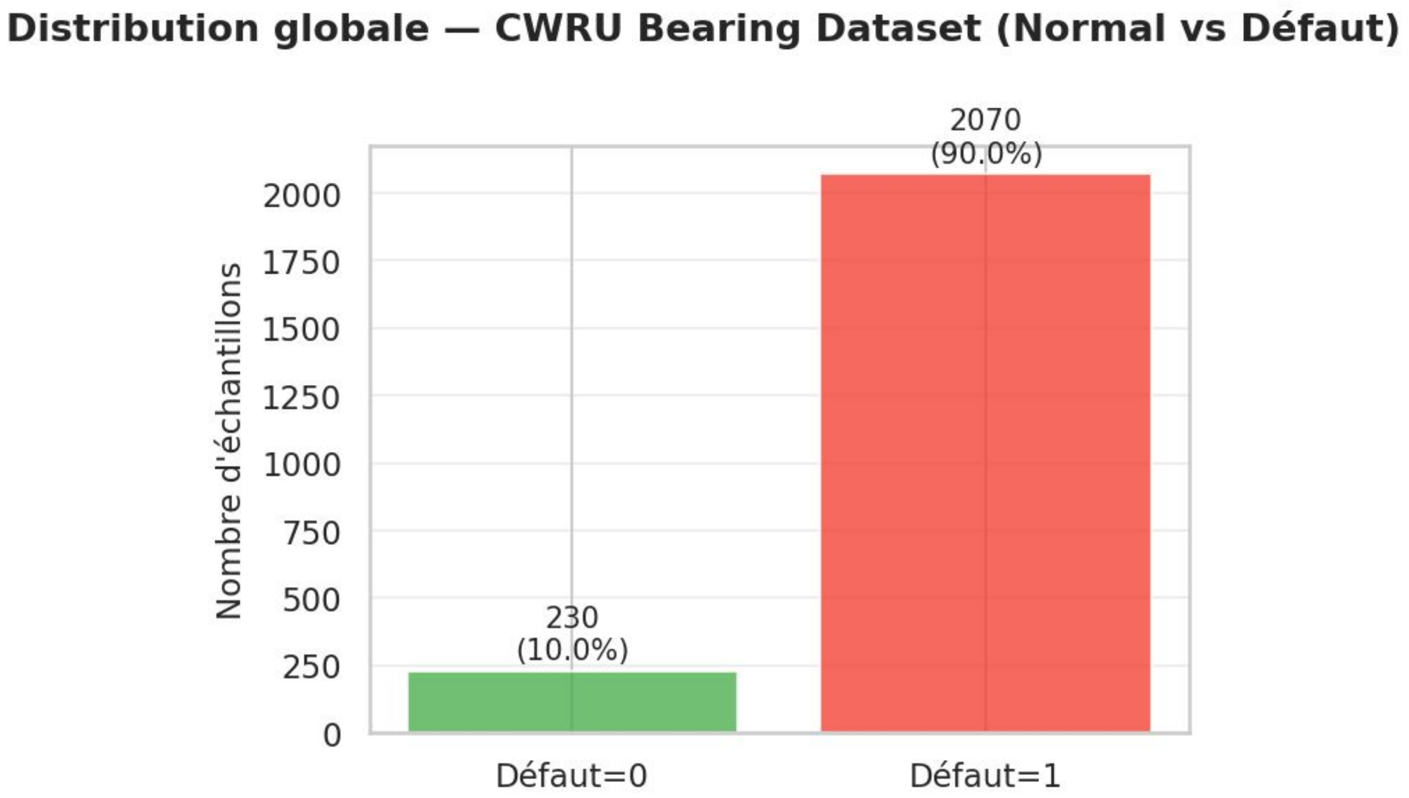
Datasets CWRU

Case Western Reserve University Bearing Dataset

Diagnostic de pannes

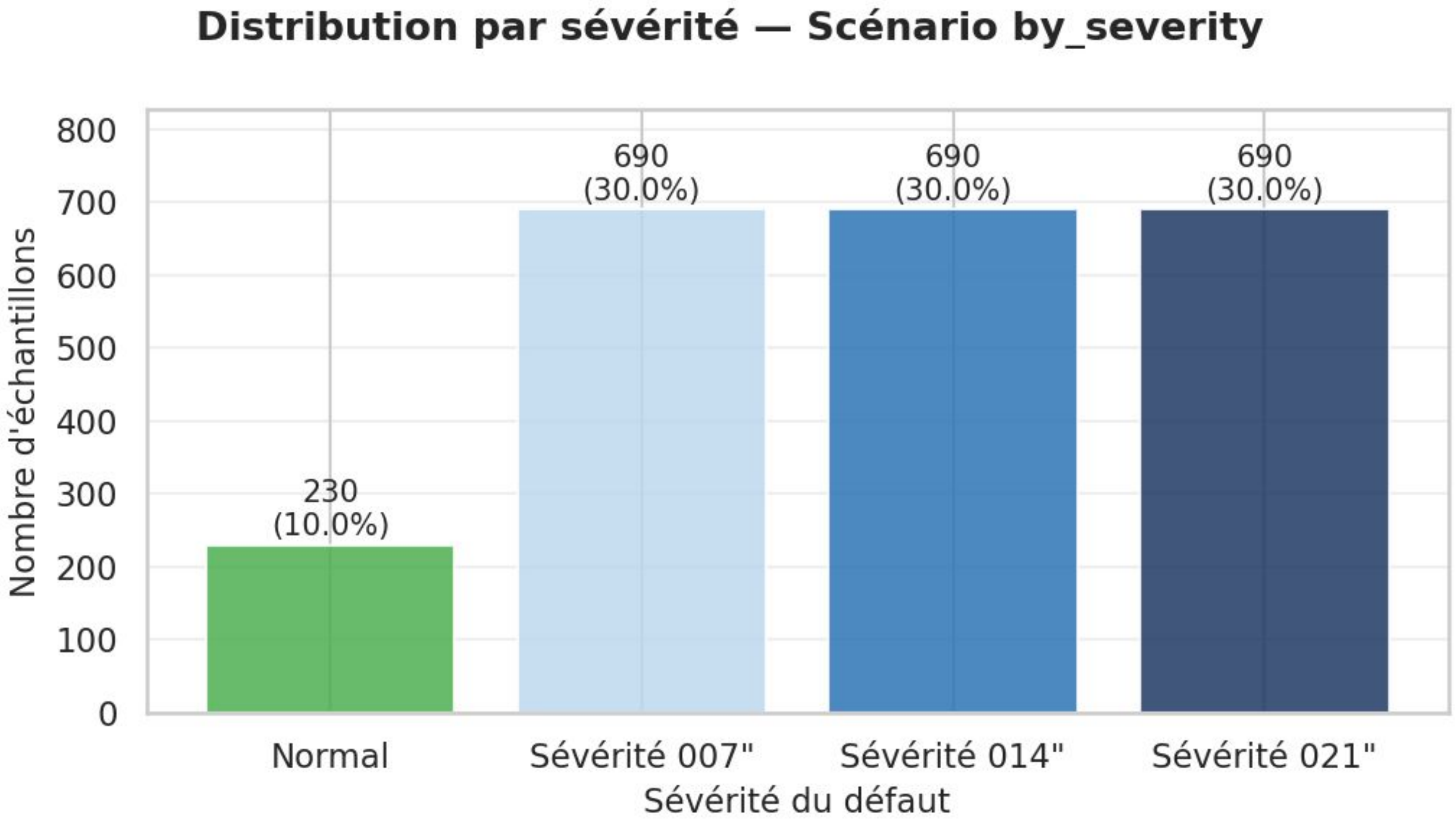
Défauts introduits artificiellement sur 3 localisation possible du roulement.

Datasets CWRU

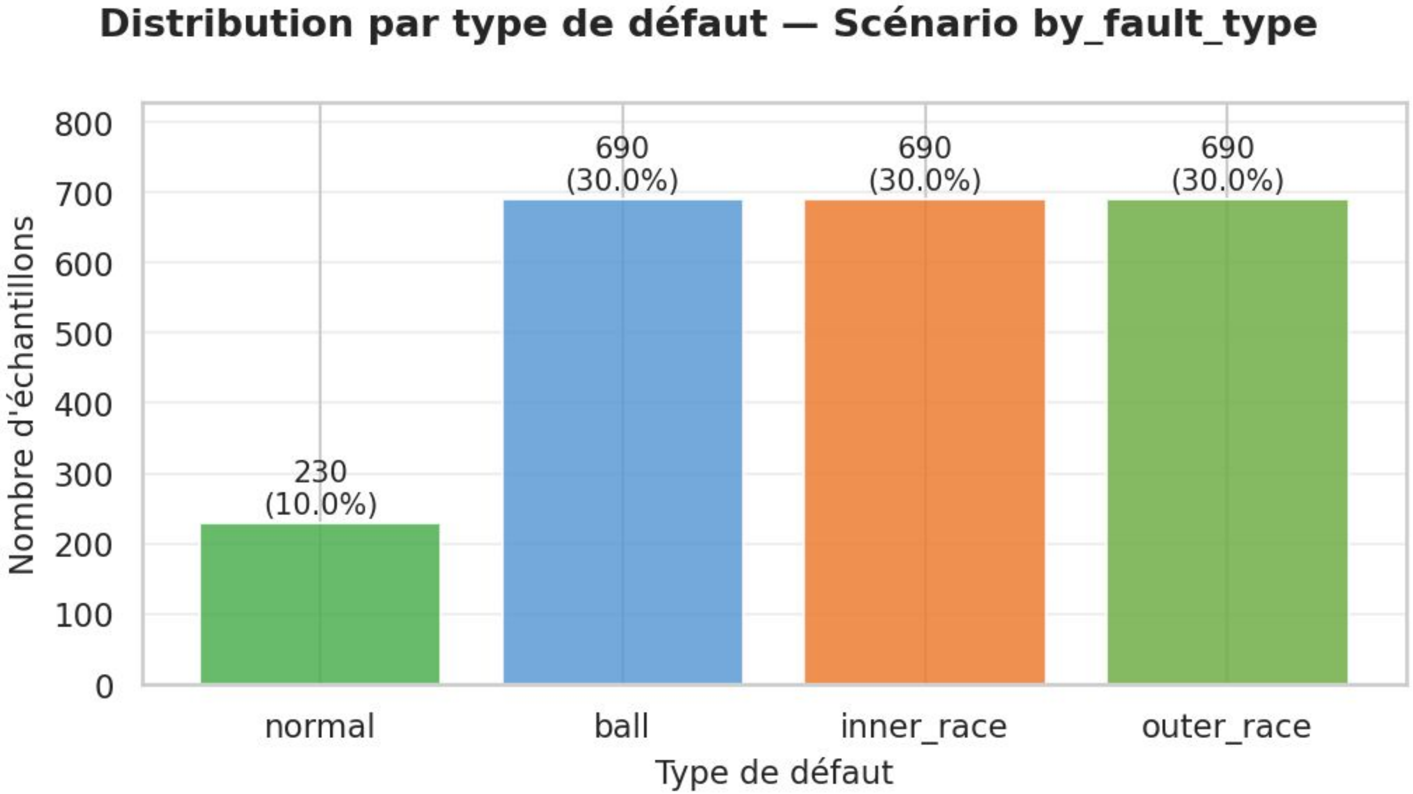


	count
fault_label	
Ball_007_1	230
Ball_014_1	230
Ball_021_1	230
IR_007_1	230
IR_014_1	230
IR_021_1	230
Normal_1	230
OR_007_6_1	230
OR_014_6_1	230
OR_021_6_1	230

Datasets CWRU Taille des défauts (0.178, 0.356, 0.533 mm)



Datasets CWRU Localisation Défaut bille (Ball), bague intérieure (Inner race), bague extérieure (Outer race)



Datasets CWRU

Vitesse de rotation 1772 rpm

Fréquence d'échantillonnage 48kHz

2300 fenêtres × 9 features, avec 48 kHz et fenêtre 2048 pts

Capteurs 2 accéléromètres Drive End et Fan End

Datasets CWRU

max : maximum de la fenêtre

min : minimum

mean : moyenne

sd : écart-type

rms : racine quadratique moyenne

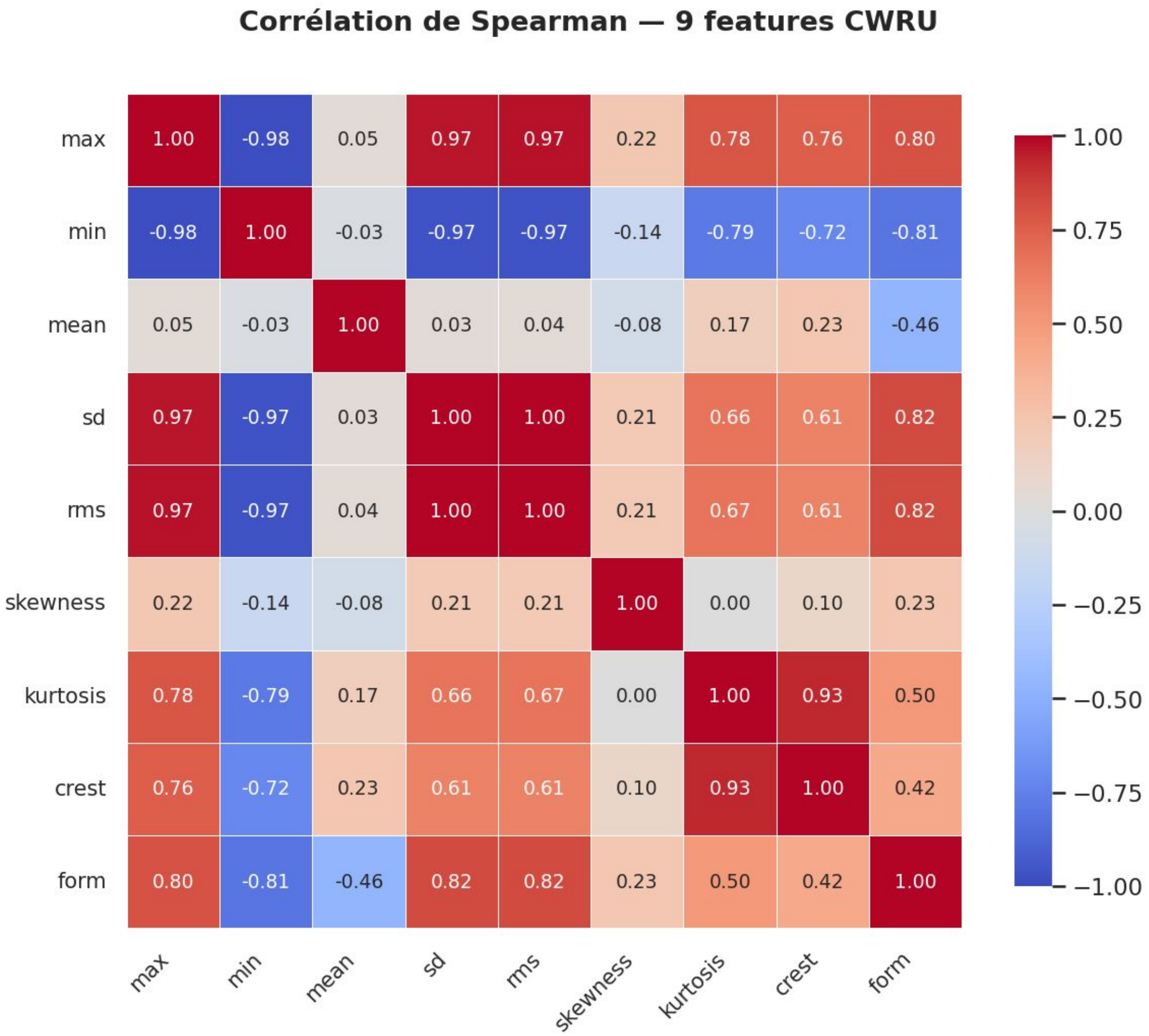
skewness : asymétrie

kurtosis : impulsivité / queues lourdes

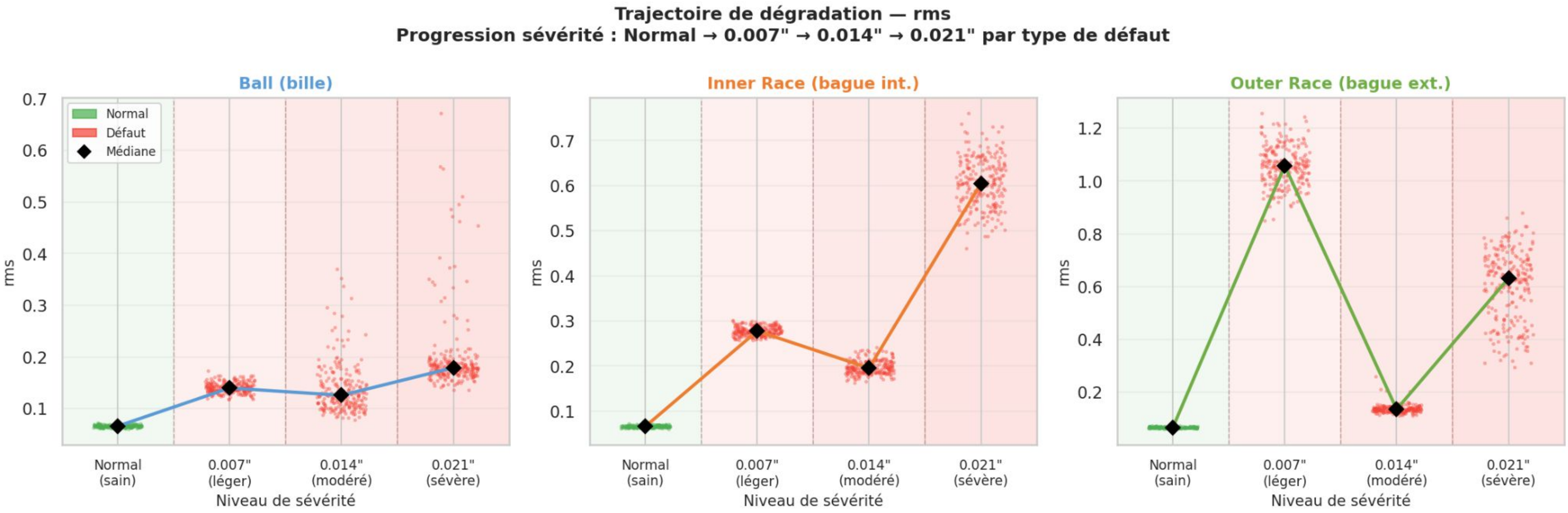
crest : facteur de crête

form : facteur de forme

Datasets CWRU



Datasets CWRU _ Trajectoire de dégradation par sévérité



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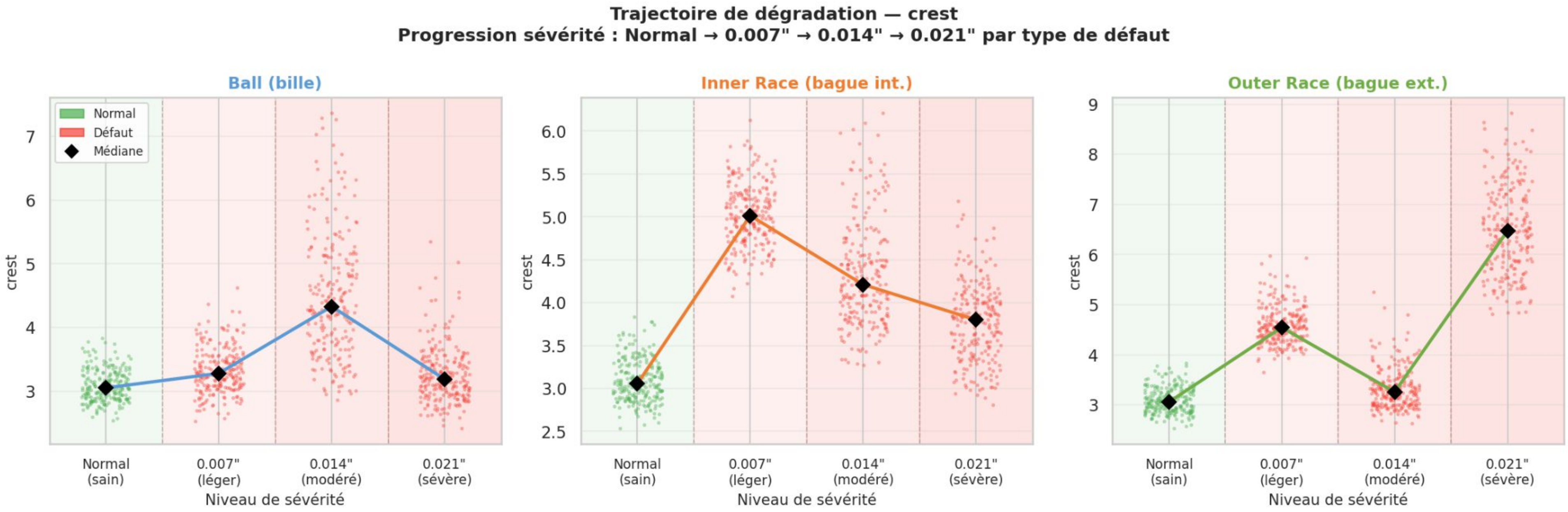
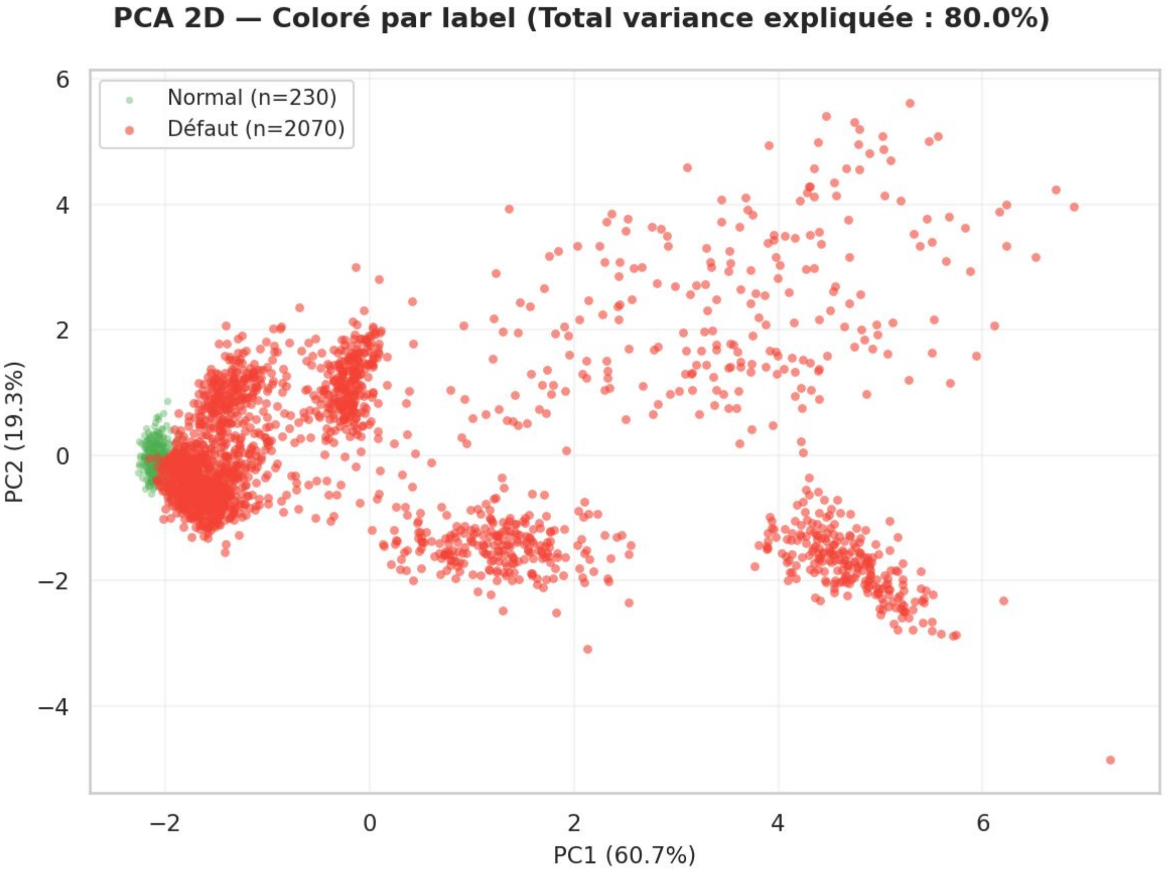
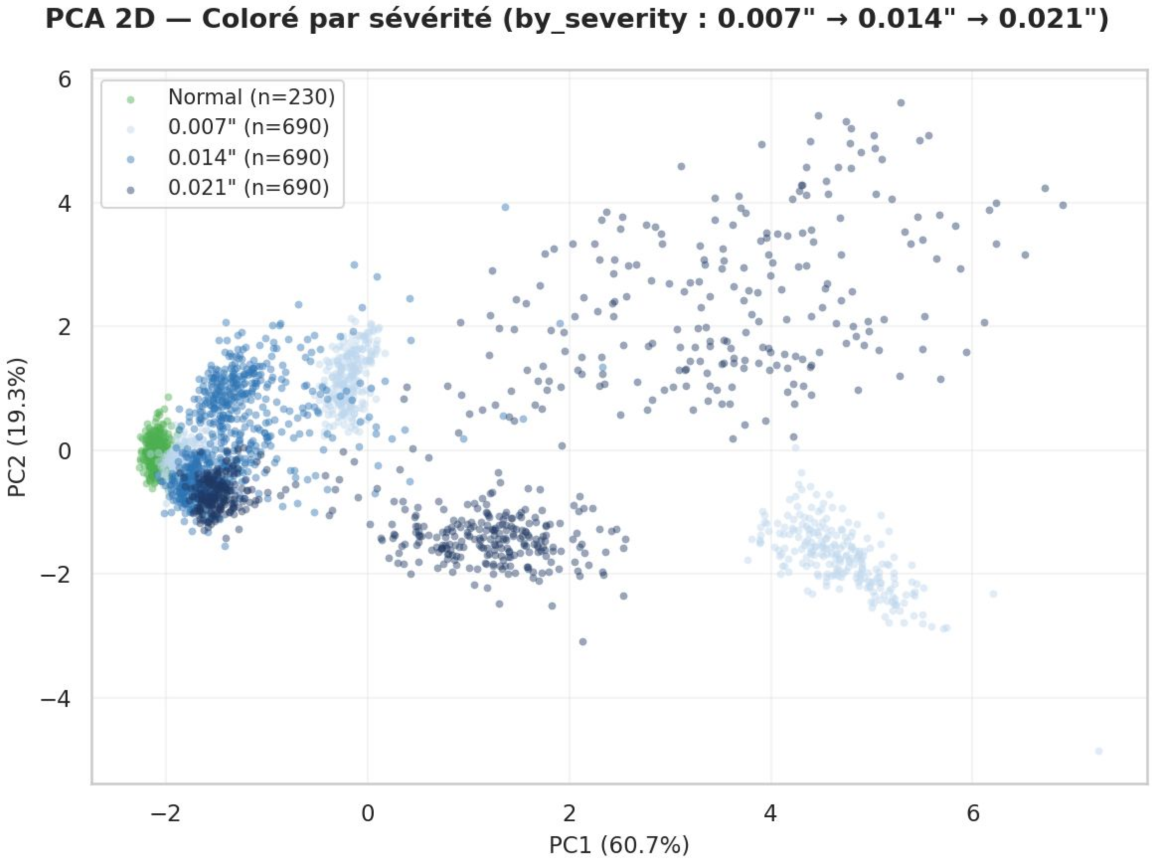


Datasets CWRU Défaut



Datasets CWRU Taille des défauts (0.178, 0.356, 0.533 mm)



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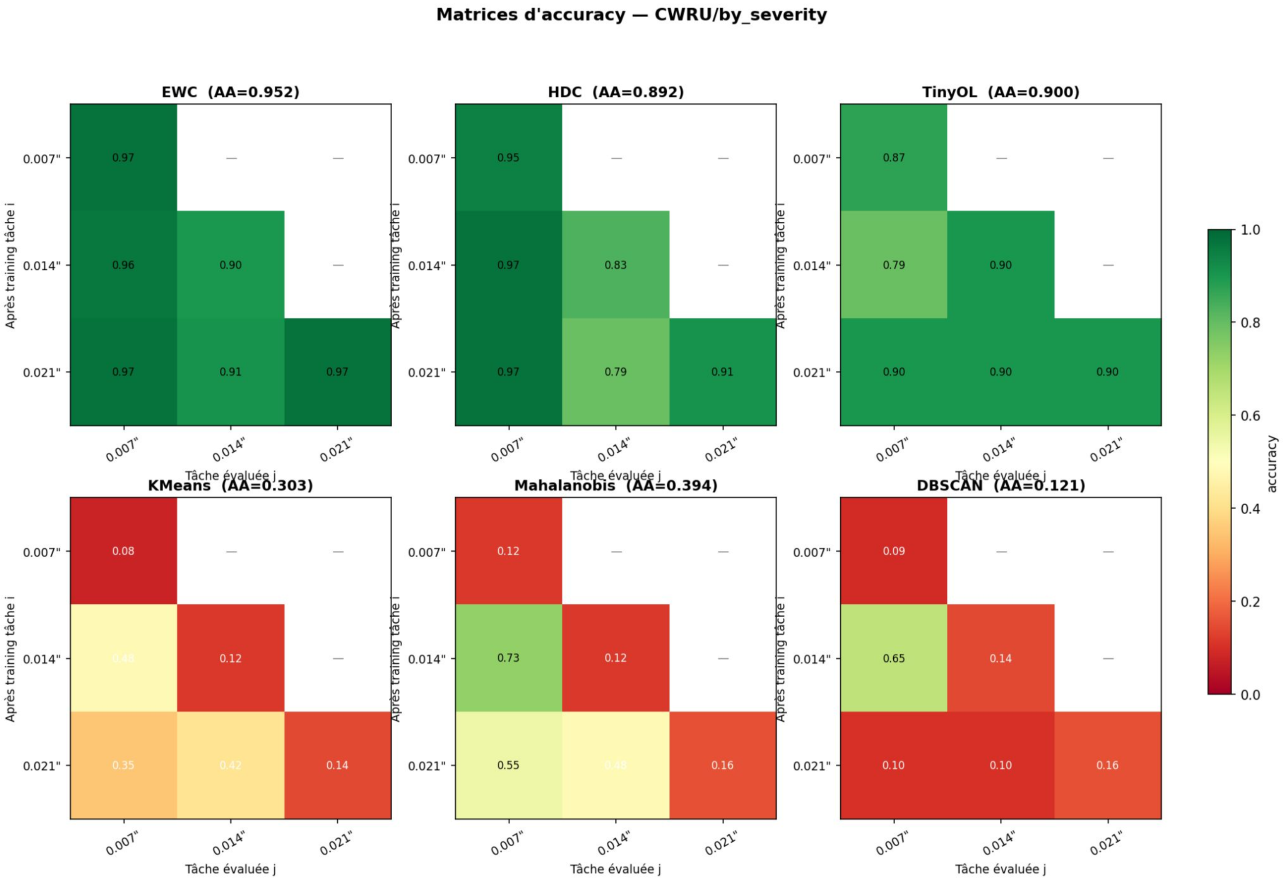

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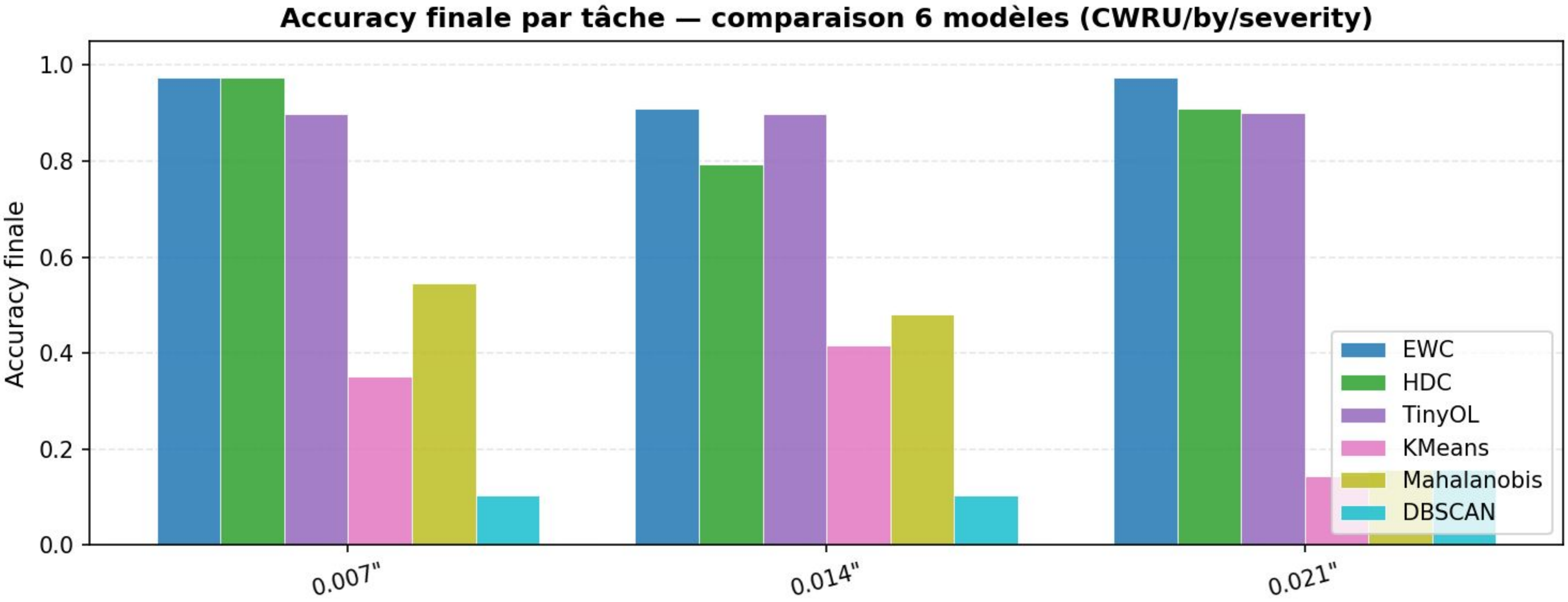

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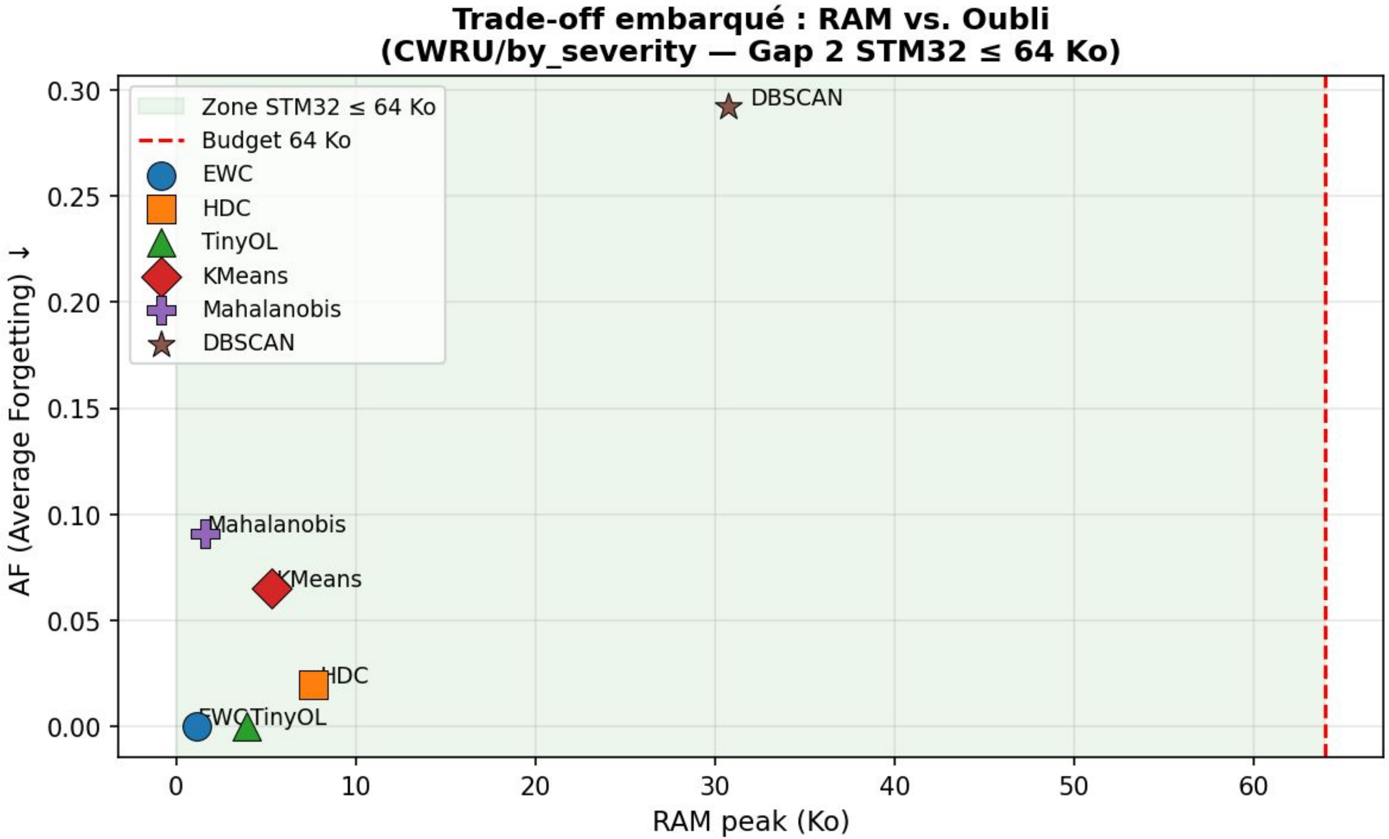

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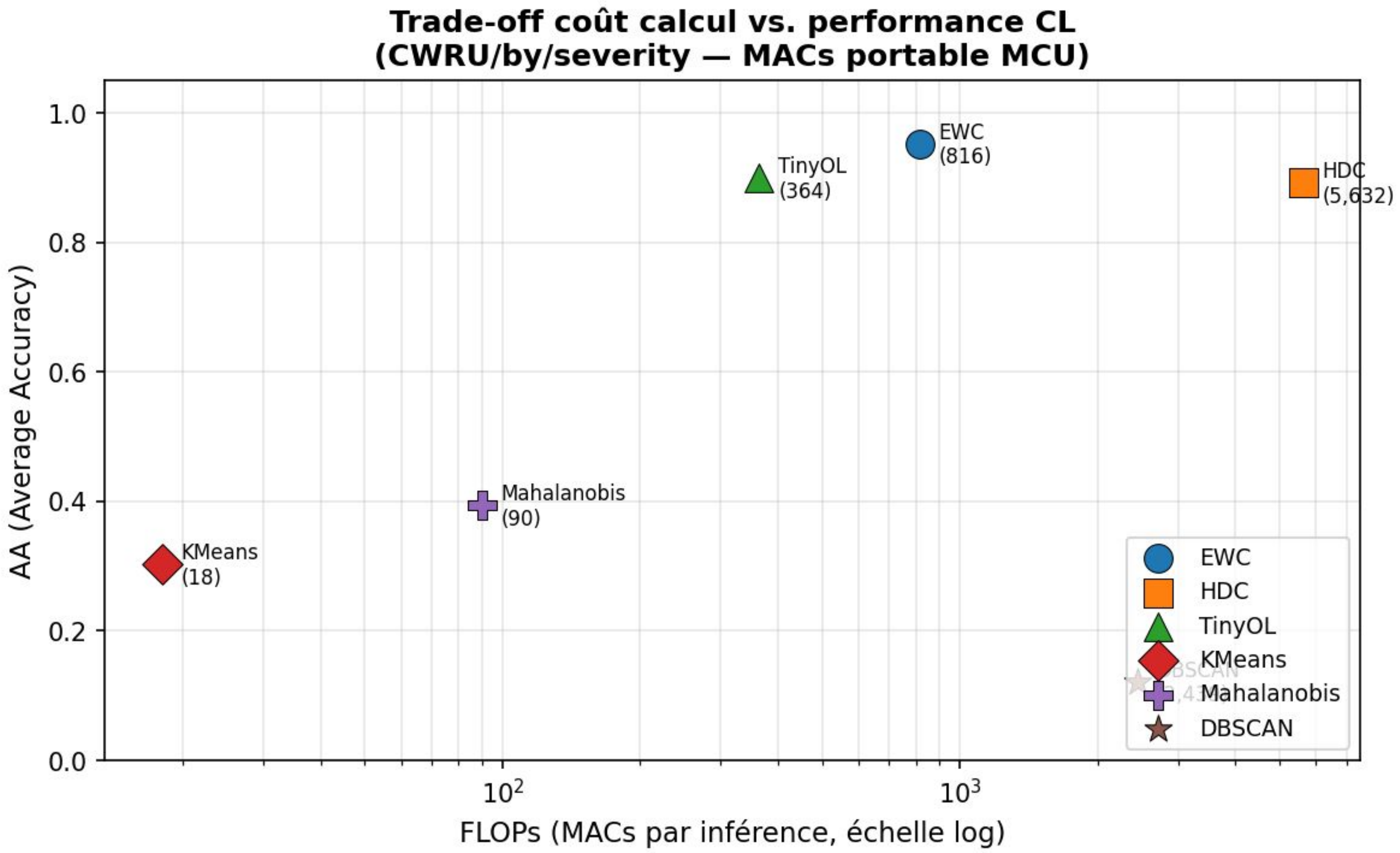

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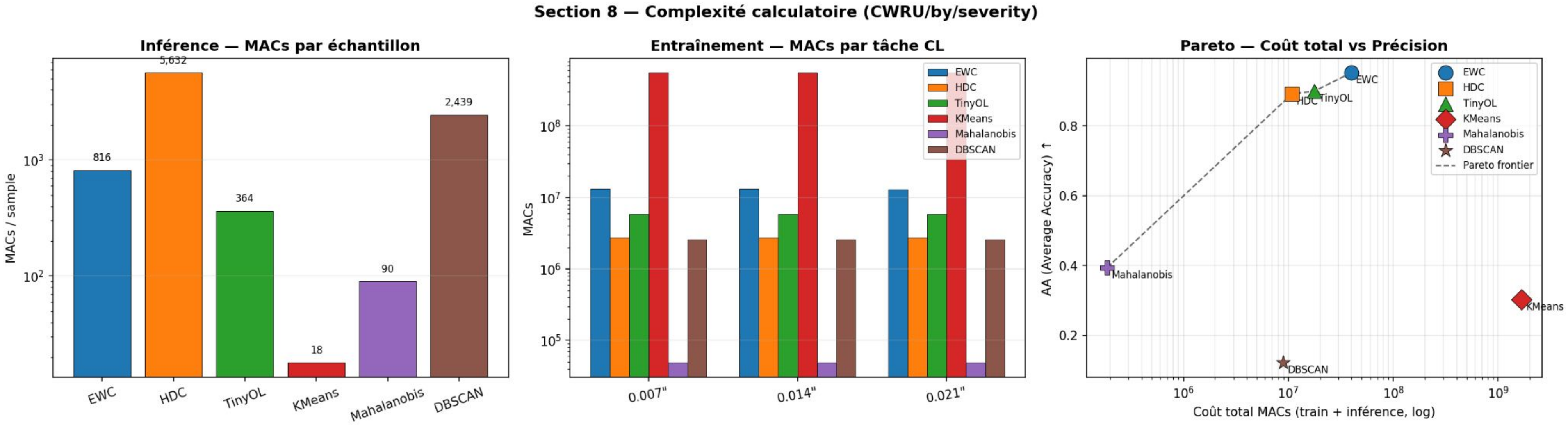
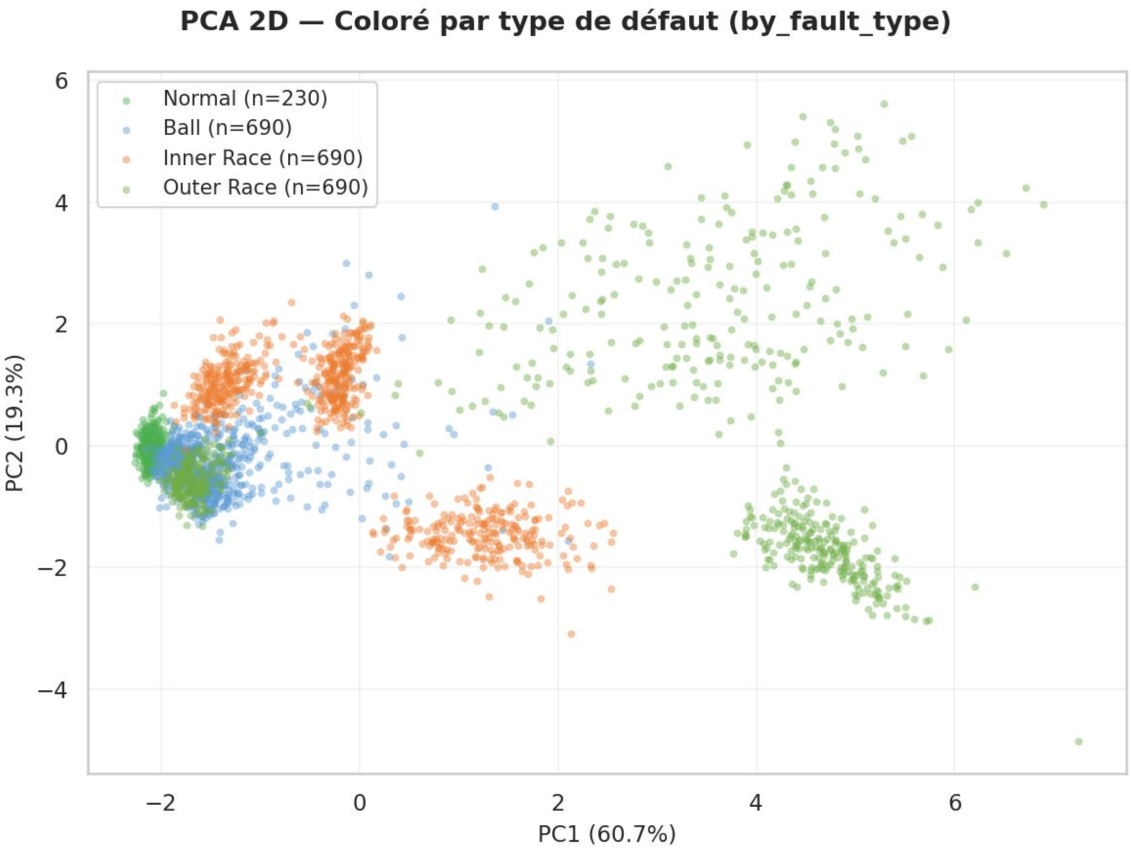


Datasets CWRU Taille des défauts (0.178, 0.356, 0.533 mm)

Datasets CWRU Type des défauts



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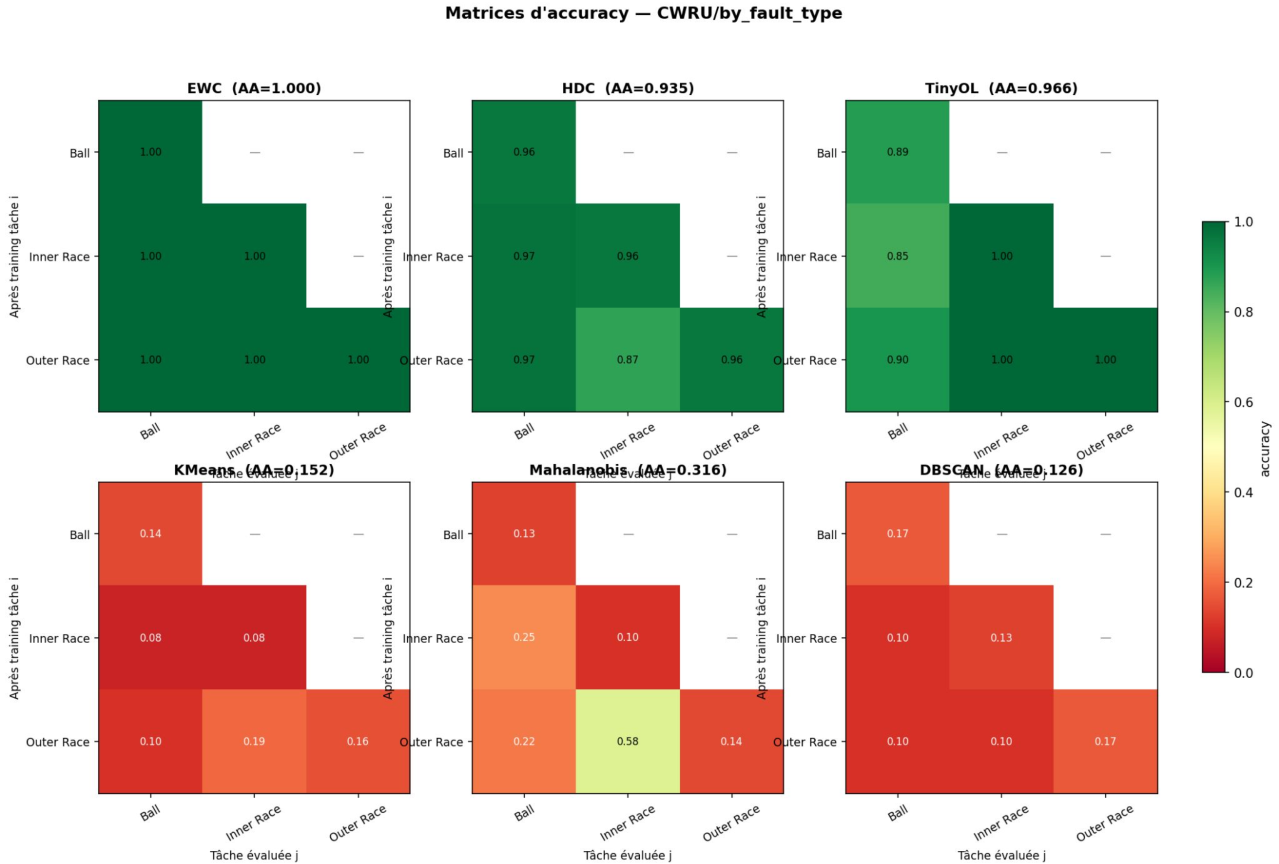

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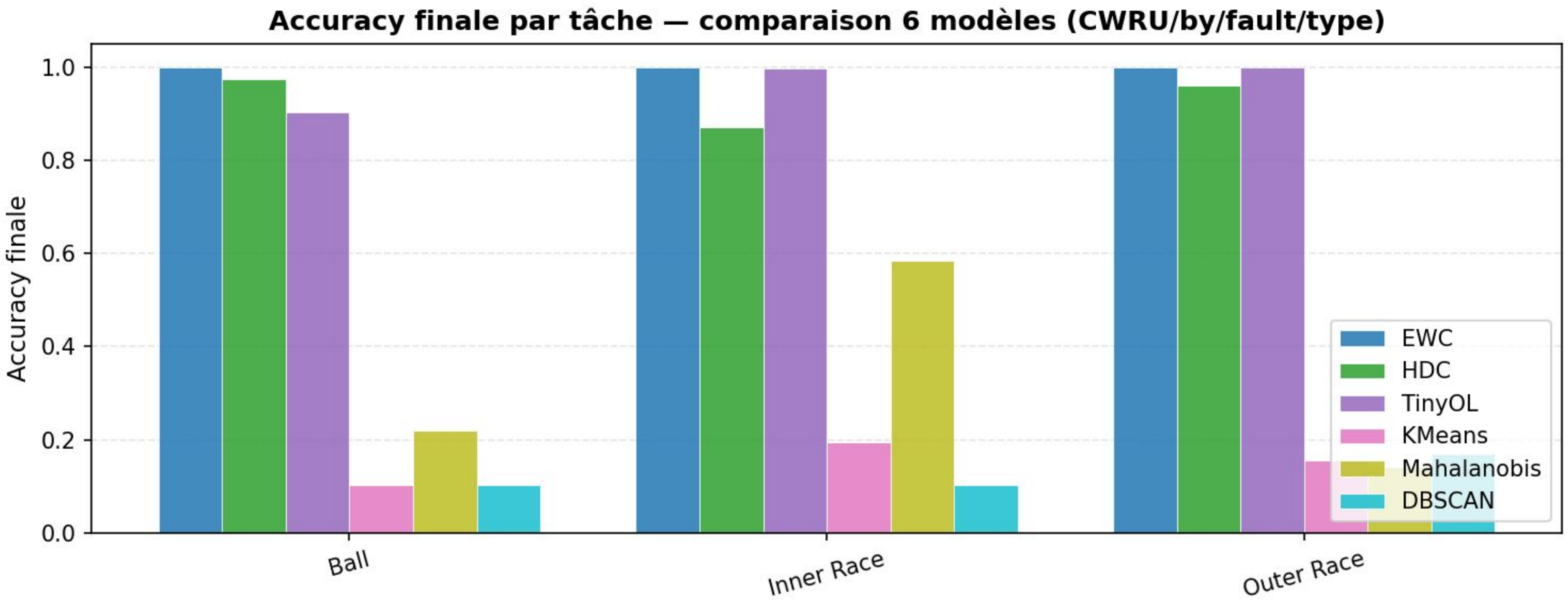

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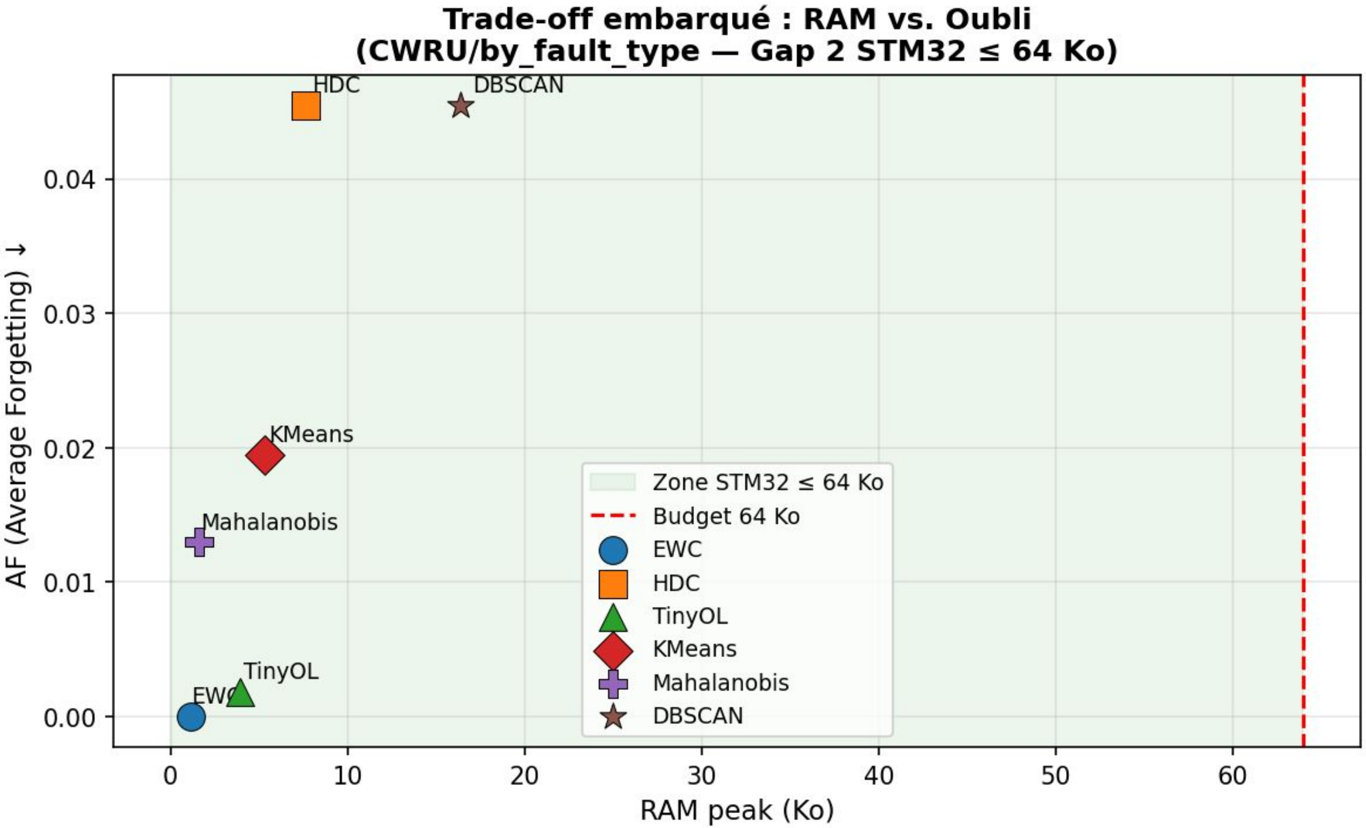

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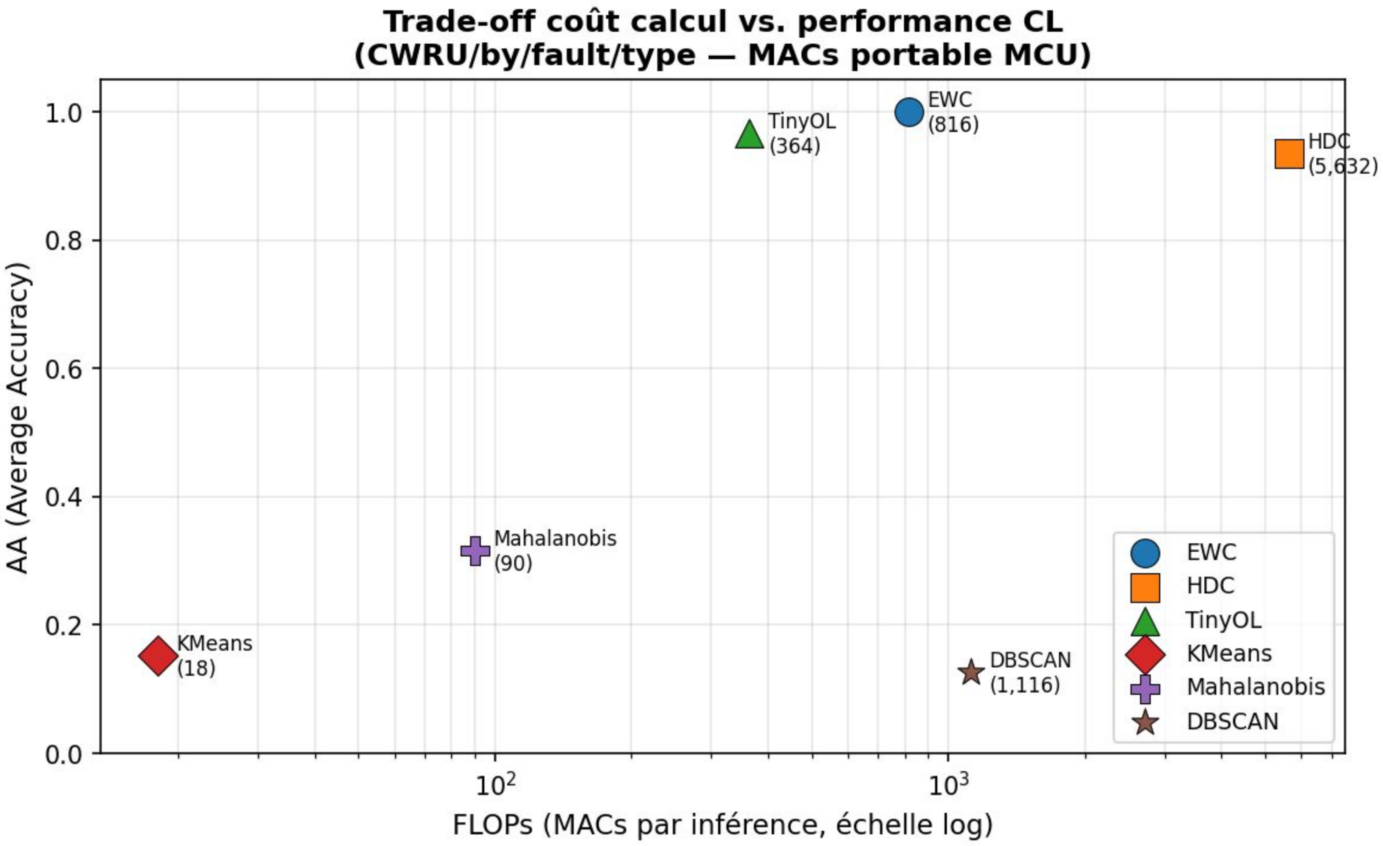

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